



Titanic Dataset — Exploratory Data Analysis

Introduction

The Titanic dataset contains information about passengers aboard the RMS Titanic.

The main objective of this exploratory data analysis (EDA) is to understand the dataset, identify patterns and relationships between passenger characteristics and survival, detect missing values and outliers, and perform feature engineering to create useful variables for further analysis.

Objectives

- Understand the structure and characteristics of the dataset
- Identify missing values
- Analyze numerical and categorical features
- Study relationships between passenger features and survival
- Detect potential outliers
- Perform feature engineering
- Summarize important findings from the analysis

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```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1. Dataset Overview

Let's begin by loading the dataset and examining its basic structure.

We will check:

- First few rows
- Number of rows and columns
- Column names
- Data types
- Basic statistical information
- Missing values

```
In [2]: titanic = pd.read_csv('Data sets/train.csv')
titanic.head()
```

```
Out[2]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cal
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	N
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	N
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	CI
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	N

```
In [3]: titanic.shape
```

```
Out[3]: (891, 12)
```

Dataset Dimensions

The dataset contains 891 observations and 12 columns.

Each row represents a passenger, while each column contains information about that passenger.

2. Understanding the Features

- PassengerId → Unique identifier for each passenger
- Survived → Survival status (0 = No, 1 = Yes) = Pclass → Passenger class (1st, 2nd, 3rd)
- Name → Passenger's name
- Sex → Passenger's gender
- Age → Passenger's age
- SibSp → Number of siblings/spouses aboard
- Parch → Number of parents/children aboard
- Ticket → Ticket number
- Fare → Passenger/ticket fare
- Cabin → Cabin number
- Embarked → Port of embarkation

Understanding these variables will help us interpret the patterns discovered during EDA.

Column Types

- **Numerical** - Age, Fare, PassengerId
- **Categorical** - Survived, Pclass, Sex, SibSp, Parch, Embarked
- **Mixed** - Name, Ticket, Cabin

3. Univariate Analysis

Univariate analysis focuses on analyzing each feature in the dataset independently.

- **Distribution analysis:** The distribution of each feature is examined to identify its shape, central tendency, and dispersion.

- **Identifying potential issues:** Univariate analysis helps in identifying potential problems with the data such as outliers, skewness, and missing values

Steps of doing Univariate Analysis on **Numerical columns**

- **Descriptive Statistics** : Compute basic summary statistics for the column, such as mean, median, mode, standard deviation, range, and quartiles. These statistics give a general understanding of the distribution of the data and can help identify skewness or outliers.
- **Visualizations** : Create visualizations to explore the distribution of the data. Some common visualizations for numerical data include histograms, box plots, and density plots. These visualizations provide a visual representation of the distribution of the data and can help identify skewness and outliers.
- **Identifying Outliers** : Identify and examine any outliers in the data. Outliers can be identified using visualizations. It is important to determine whether the outliers are due to measurement errors, data entry errors, or legitimate differences in the data, and to decide whether to include or exclude them from the analysis.
- **Skewness** : Check for skewness in the data and consider transforming the data or using robust statistical methods that are less sensitive to skewness, if necessary.
- **Conclusion** : Summarize the findings of the EDA and make decisions about how to proceed with further analysis.

3.1 Numerical Features

Age

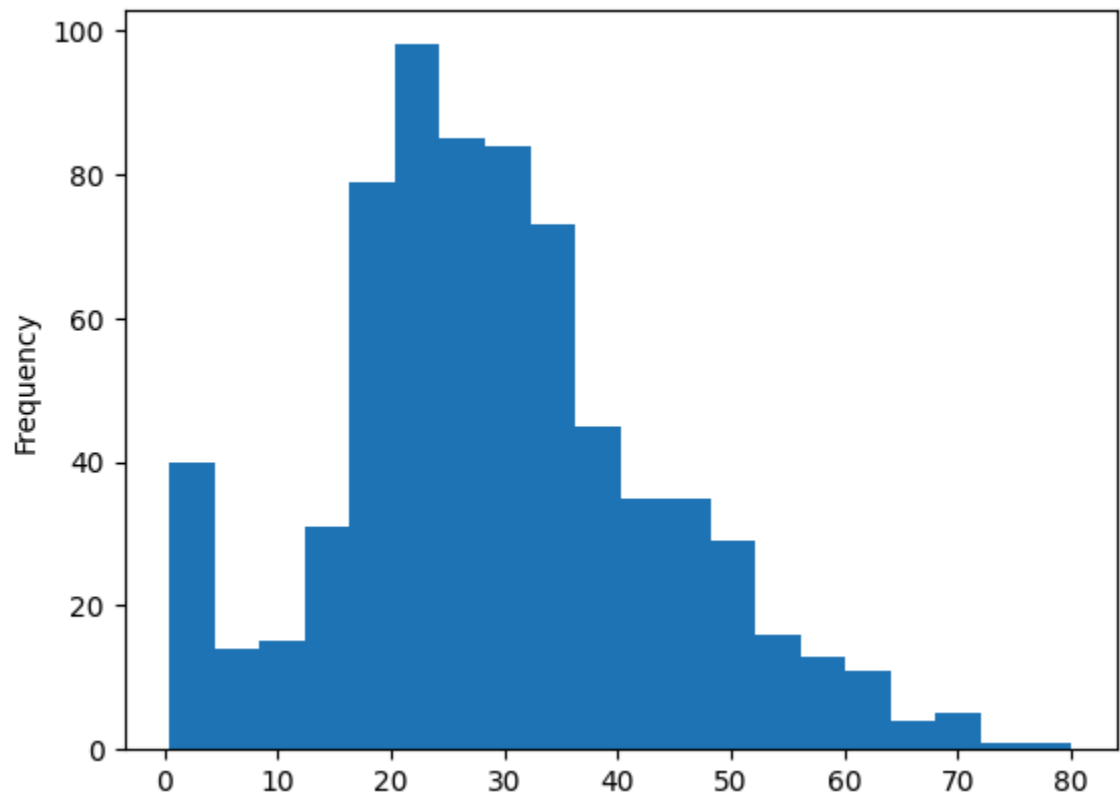
Conclusions

- Age ranges from approximately 0.4 to 80 years.
- Around 20% of the Age values are missing.
- The distribution is broadly concentrated around young and middle-aged passengers.

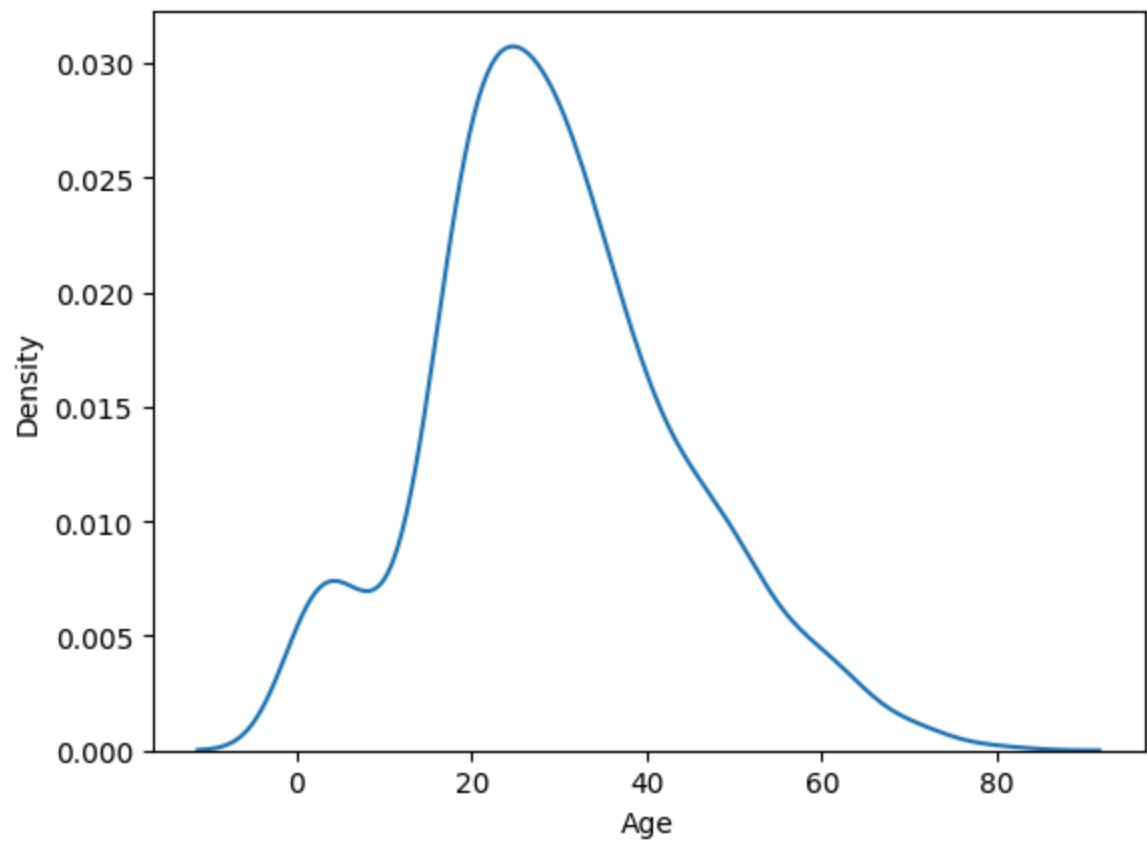
```
In [4]: titanic['Age'].describe()
```

```
Out[4]: count    714.000000  
mean      29.699118  
std       14.526497  
min        0.420000  
25%       20.125000  
50%       28.000000  
75%       38.000000  
max       80.000000  
Name: Age, dtype: float64
```

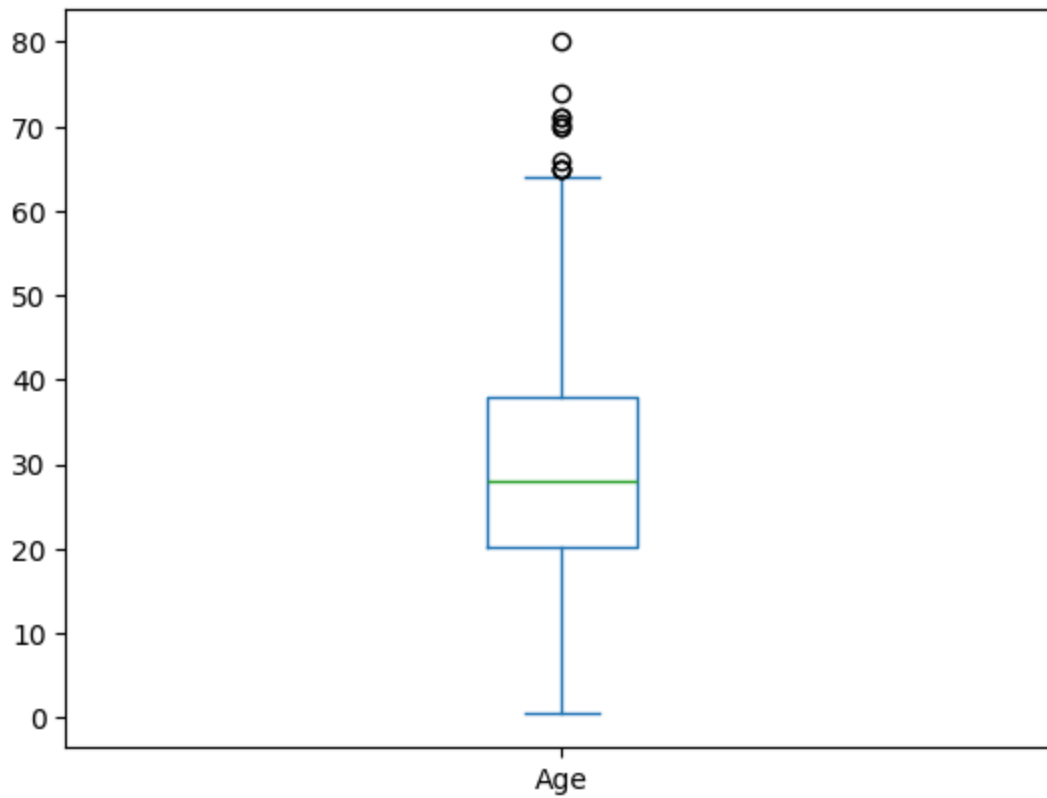
```
In [5]: titanic['Age'].plot(kind='hist', bins=20);
```



```
In [6]: sns.kdeplot(data=titanic, x='Age');
```



```
In [7]: titanic['Age'].plot(kind='box');
```



```
In [8]: # check for outliers and print them
titanic[titanic['Age'] > 65]
```

Out[8]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
33	34	0	2	Wheadon, Mr. Edward H	male	66.0	0	0	C.A. 24579	10.5000	
96	97	0	1	Goldschmidt, Mr. George B	male	71.0	0	0	PC 17754	34.6542	
116	117	0	3	Connors, Mr. Patrick	male	70.5	0	0	370369	7.7500	
493	494	0	1	Artagaveytia, Mr. Ramon	male	71.0	0	0	PC 17609	49.5042	
630	631	1	1	Barkworth, Mr. Algernon Henry Wilson	male	80.0	0	0	27042	30.0000	
672	673	0	2	Mitchell, Mr. Henry Michael	male	70.0	0	0	C.A. 24580	10.5000	
745	746	0	1	Crosby, Capt. Edward Gifford	male	70.0	1	1	WE/P 5735	71.0000	
851	852	0	3	Svensson, Mr. Johan	male	74.0	0	0	347060	7.7750	

```
In [9]: (titanic['Age'].isnull().sum() / len(titanic['Age']) ) * 100
```

```
Out[9]: np.float64(19.865319865319865)
```

Fare

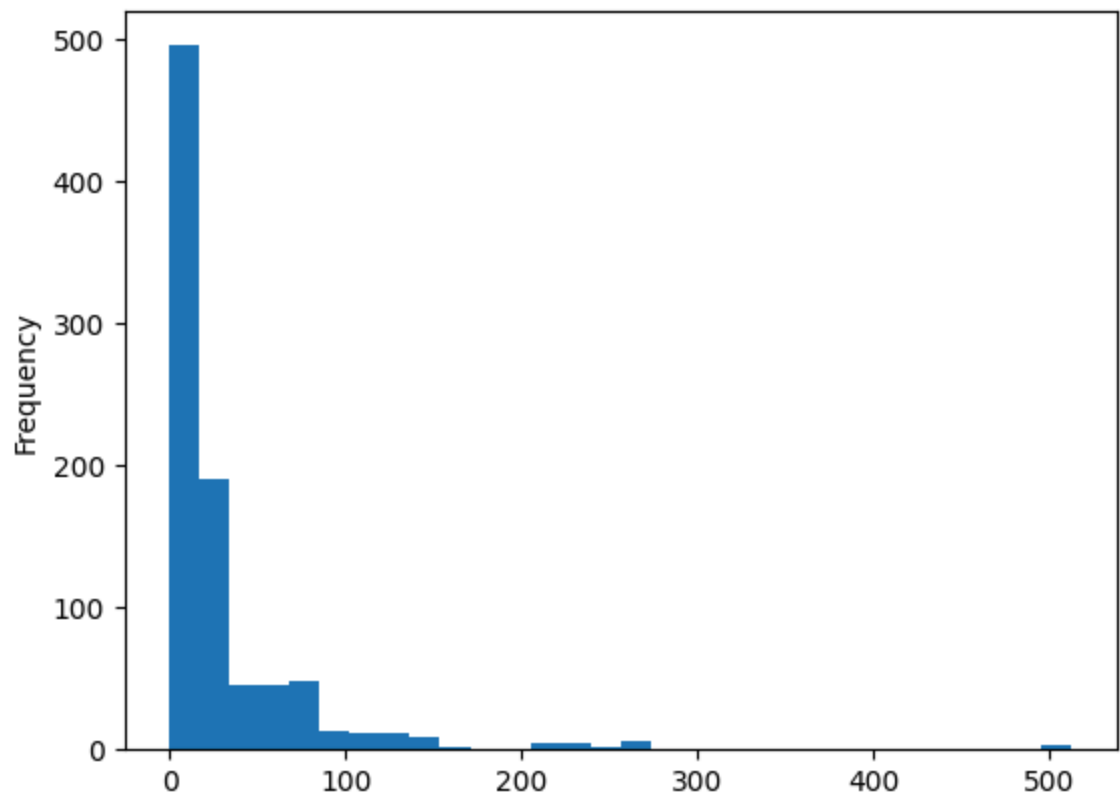
Conclusions

- Data is highly postively skewed.
- The `Fare` column represents the fare associated with the ticket. When multiple passengers share the same ticket, this amount may represent the fare for the group rather than the amount attributable to one passenger.
- To obtain an approximate fare per passenger, we divide the ticket fare by the number of passengers associated with the ticket.

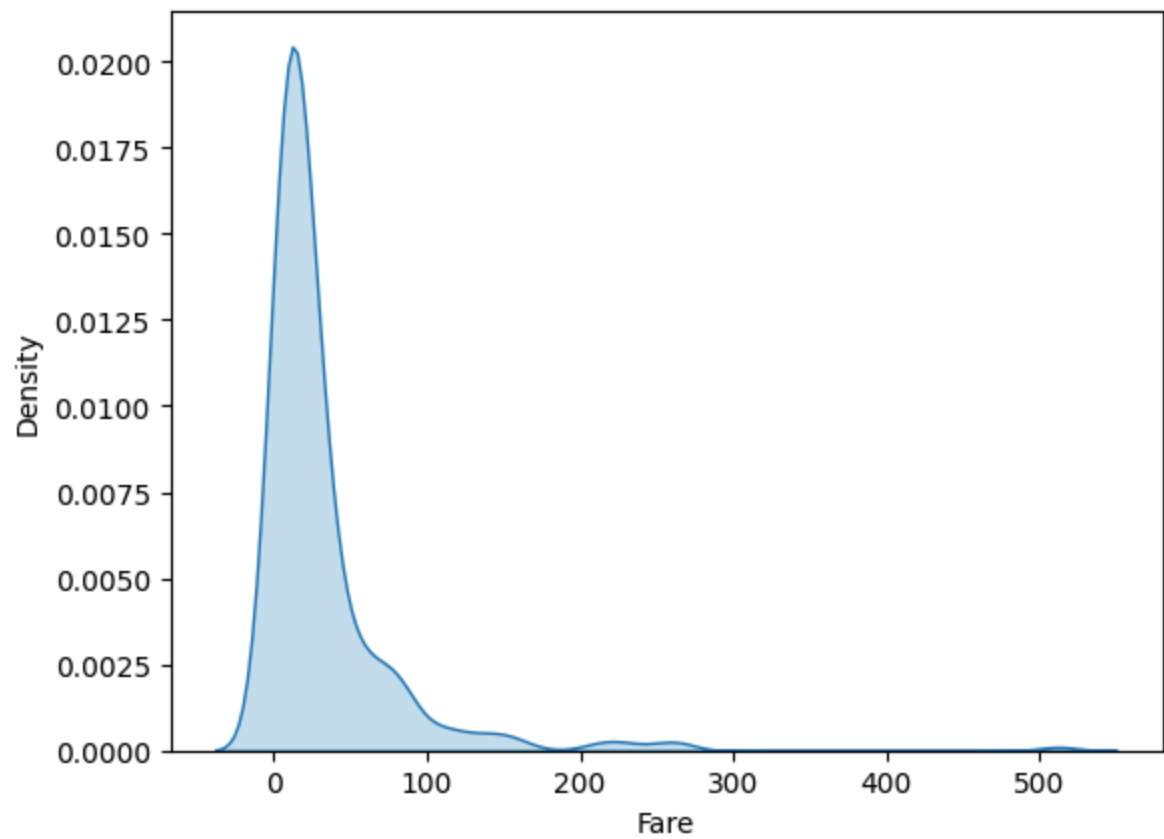
```
In [10]: titanic['Fare'].describe()
```

```
Out[10]: count      891.000000  
mean         32.204208  
std          49.693429  
min           0.000000  
25%          7.910400  
50%         14.454200  
75%         31.000000  
max         512.329200  
Name: Fare, dtype: float64
```

```
In [11]: titanic['Fare'].plot(kind='hist', bins=30);
```



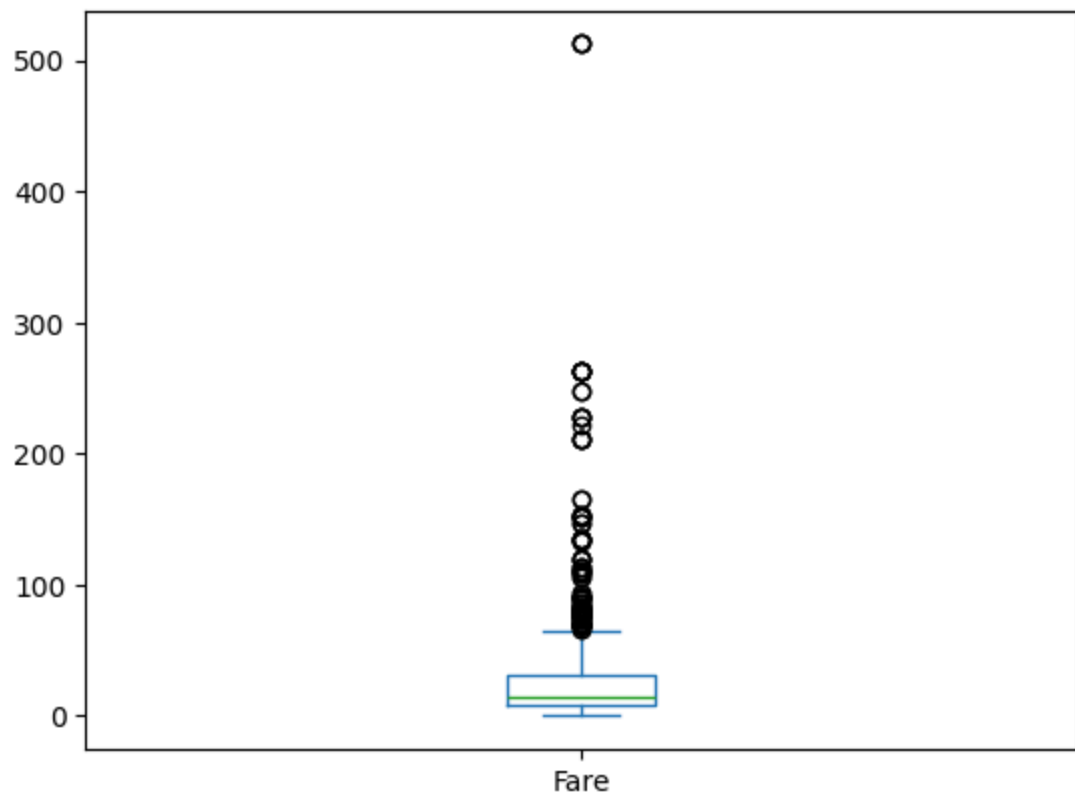
In [12... `sns.kdeplot(data=titanic, x='Fare', fill=True);`



```
In [13... titanic['Fare'].skew()
```

```
Out[13]: np.float64(4.787316519674893)
```

```
In [14... titanic['Fare'].plot(kind='box');
```



```
In [15... # check the outliers  
titanic[titanic['Fare'] > 250]
```

Out[15]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
	27	0	1	Fortune, Mr. Charles Alexander	male	19.0	3	2	19950	263.0000
	88	1	1	Fortune, Miss. Mabel Helen	female	23.0	3	2	19950	263.0000
	258	1	1	Ward, Miss. Anna	female	35.0	0	0	PC 17755	512.3292
	311	1	1	Ryerson, Miss. Emily Borie	female	18.0	2	2	PC 17608	262.3750
	341	1	1	Fortune, Miss. Alice Elizabeth	female	24.0	3	2	19950	263.0000
	438	0	1	Fortune, Mr. Mark	male	64.0	1	4	19950	263.0000
	679	1	1	Cardeza, Mr. Thomas Drake Martinez	male	36.0	0	1	PC 17755	512.3292
	737	1	1	Lesurer, Mr. Gustave J	male	35.0	0	0	PC 17755	512.3292
	742	1	1	Ryerson, Miss. Susan Parker "Suzette"	female	21.0	2	2	PC 17608	262.3750

```
In [16... titanic['Fare'].isnull().sum()
```

Out[16]: np.int64(0)

Steps of doing Univariate Analysis on **Categorical columns**

Descriptive Statistics : Compute the frequency distribution of the categories in the column. This will give a general understanding of the distribution of the categories and their relative frequencies.

Visualizations : Create visualizations to explore the distribution of the

categories. Some common visualizations for categorical data include count plots and pie charts. These visualizations provide a visual representation of the distribution of the categories and can help identify any patterns or anomalies in the data.

Missing Values : Check for missing values in the data and decide how to handle them. Missing values can be imputed or excluded from the analysis, depending on the research question and the data set.

Conclusion : Summarize the findings of the EDA and make decisions about how to proceed with further analysis.

3.2 Categorical Features

Survived

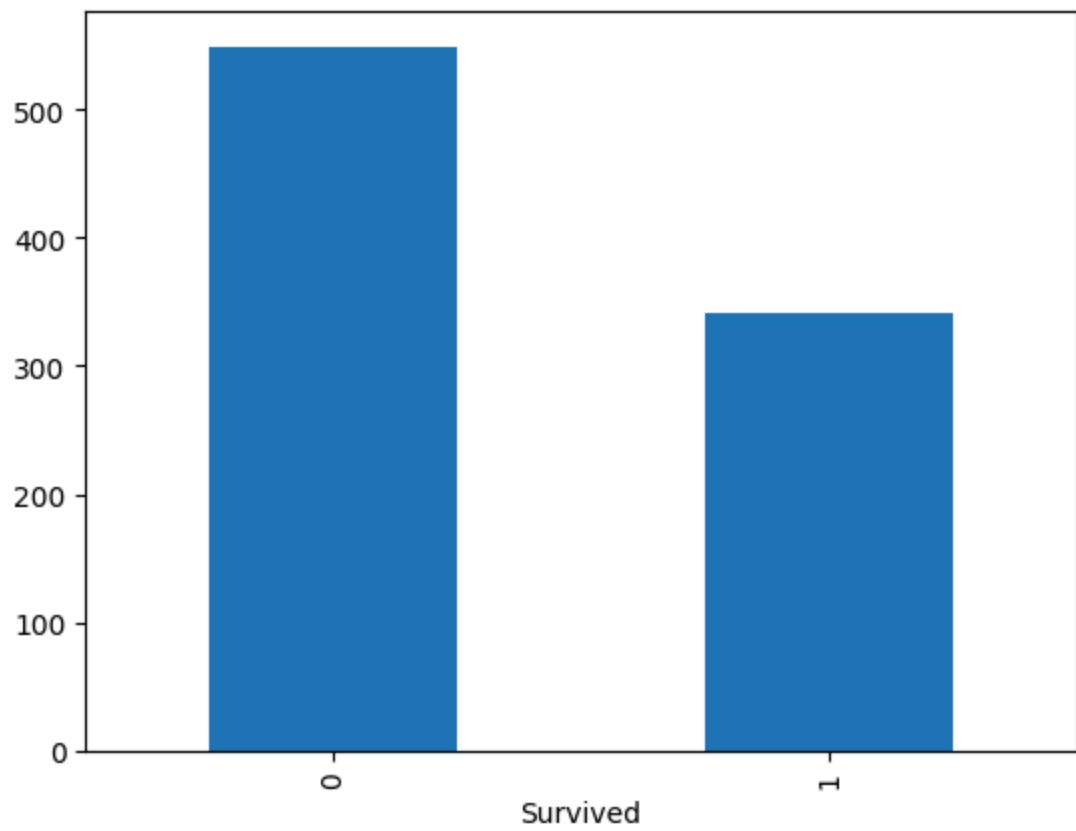
Conclusion

- Approximately 61.6% of passengers in the dataset did not survive, while 38.4% survived.

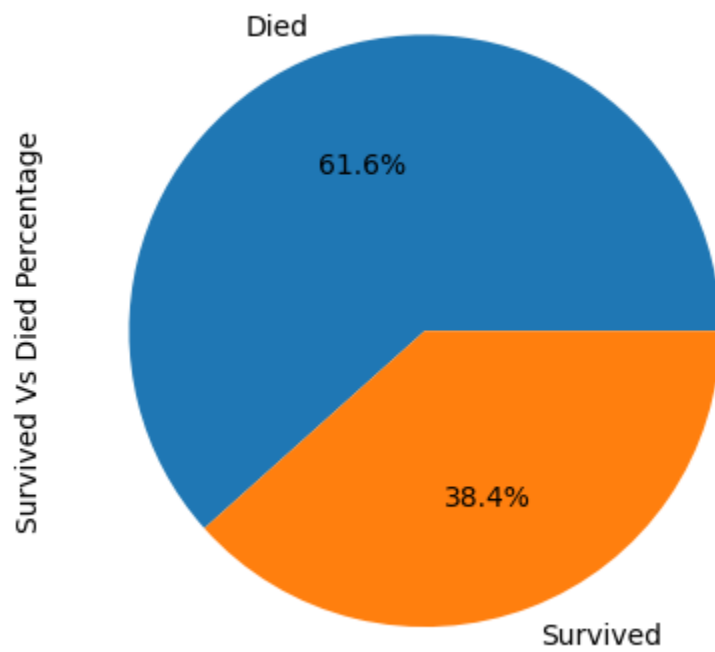
```
In [17...] titanic['Survived'].value_counts()
```

```
Out[17]: Survived
0      549
1      342
Name: count, dtype: int64
```

```
In [18...] titanic['Survived'].value_counts().plot(kind='bar');
```



In [19... `titanic['Survived'].value_counts().plot(kind='pie', autopct='%0.1f%%', la`



```
In [20]: titanic['Survived'].isnull().sum()
```

```
Out[20]: np.int64(0)
```

Pclass

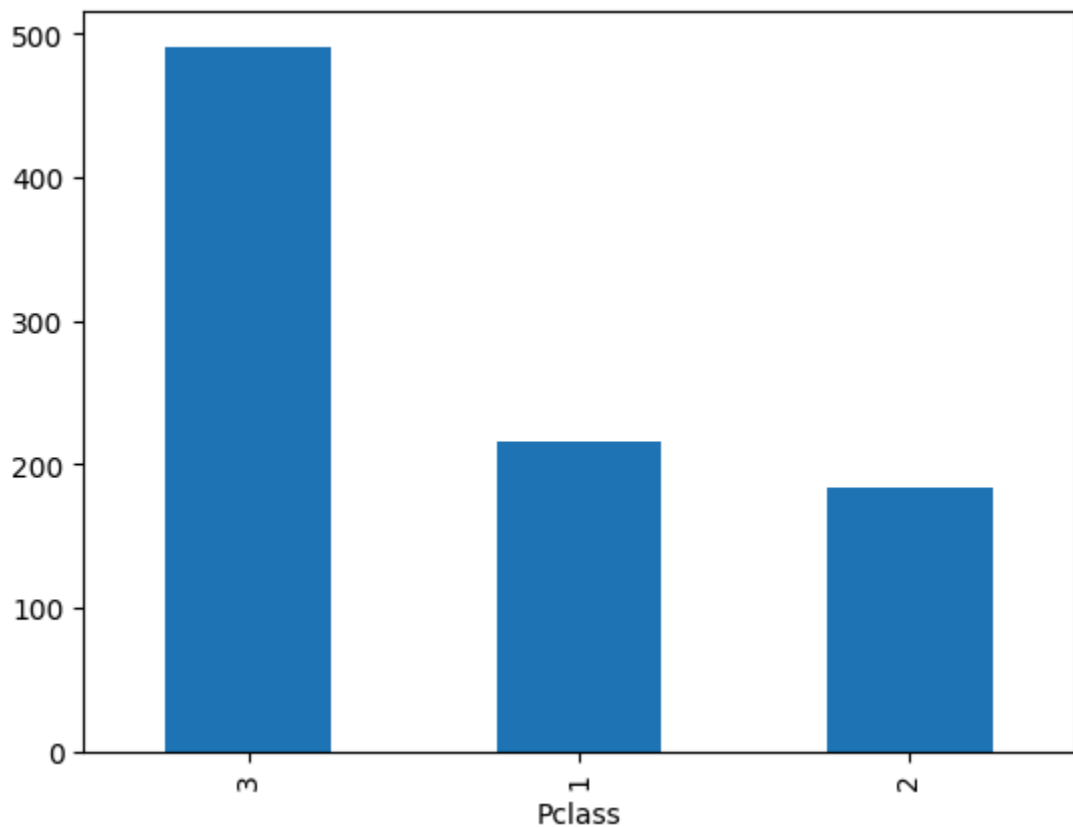
Conclusion

- The number of passengers in Pclass 1 is 216.
- The number of passengers in Pclass 2 is 184.
- The number of passengers in Pclass 3 is 491.

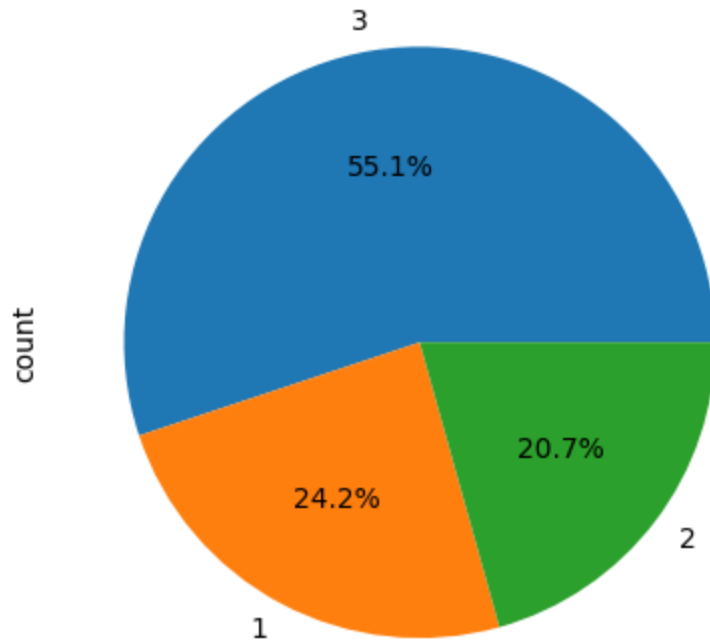
```
In [21]: titanic['Pclass'].value_counts()
```

```
Out[21]: Pclass
3      491
1      216
2      184
Name: count, dtype: int64
```

```
In [22]: titanic['Pclass'].value_counts().plot(kind='bar');
```



```
In [23]: titanic['Pclass'].value_counts().plot(kind='pie', autopct='%0.1f%%');
```



```
In [24...] titanic['Survived'].isnull().sum()
```

```
Out[24]: np.int64(0)
```

Sex

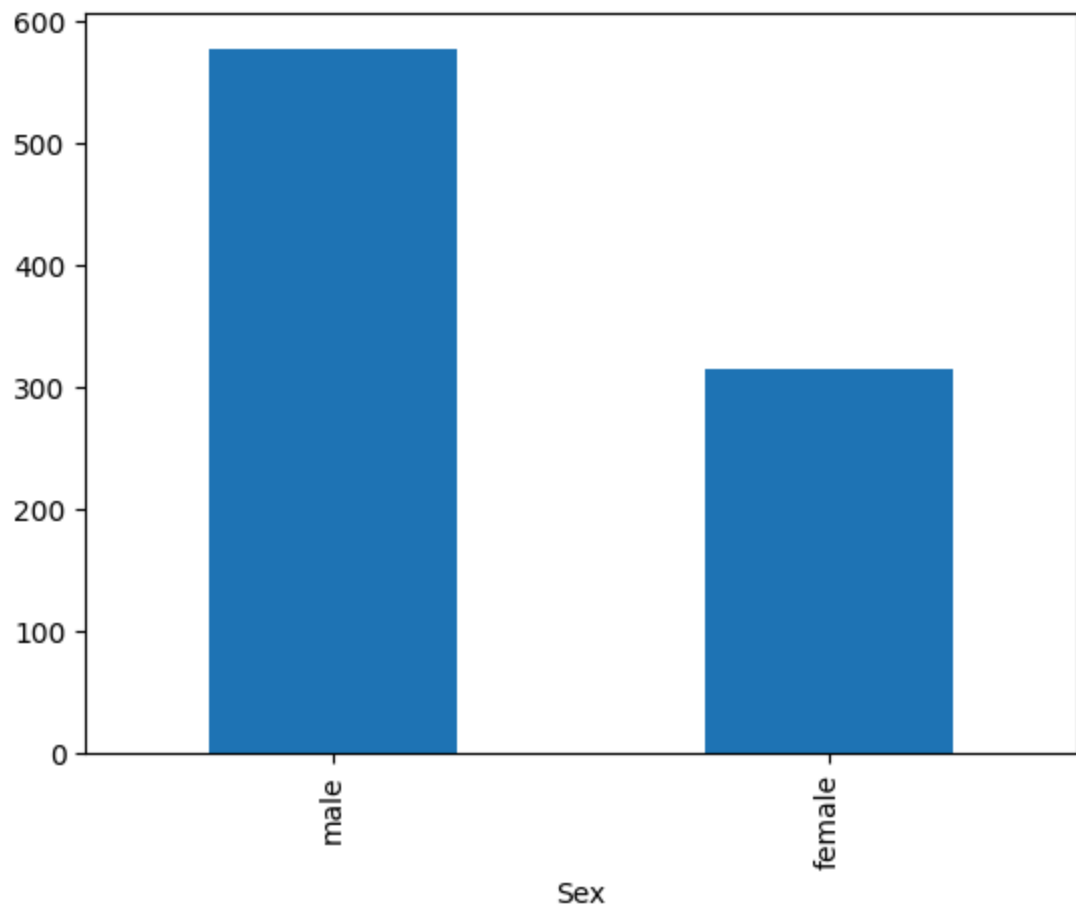
Conclusion

- There are 64.8% Males.
- There are 35.2% Females.

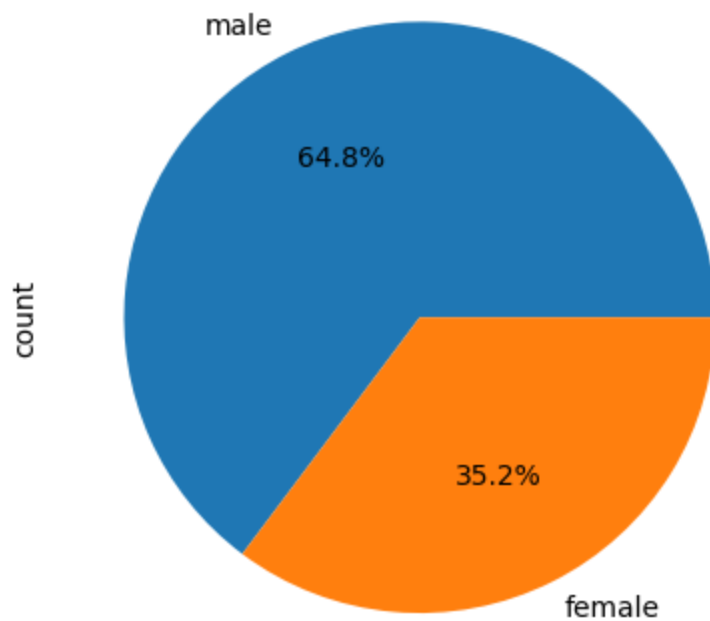
```
In [25...] titanic['Sex'].value_counts()
```

```
Out[25]: Sex
male      577
female    314
Name: count, dtype: int64
```

```
In [26...] titanic['Sex'].value_counts().plot(kind='bar');
```



In [27... `titanic['Sex'].value_counts().plot(kind='pie', autopct='%0.1f%%');`



```
In [28... titanic['Sex'].isnull().sum()
```

```
Out[28]: np.int64(0)
```

SibSp

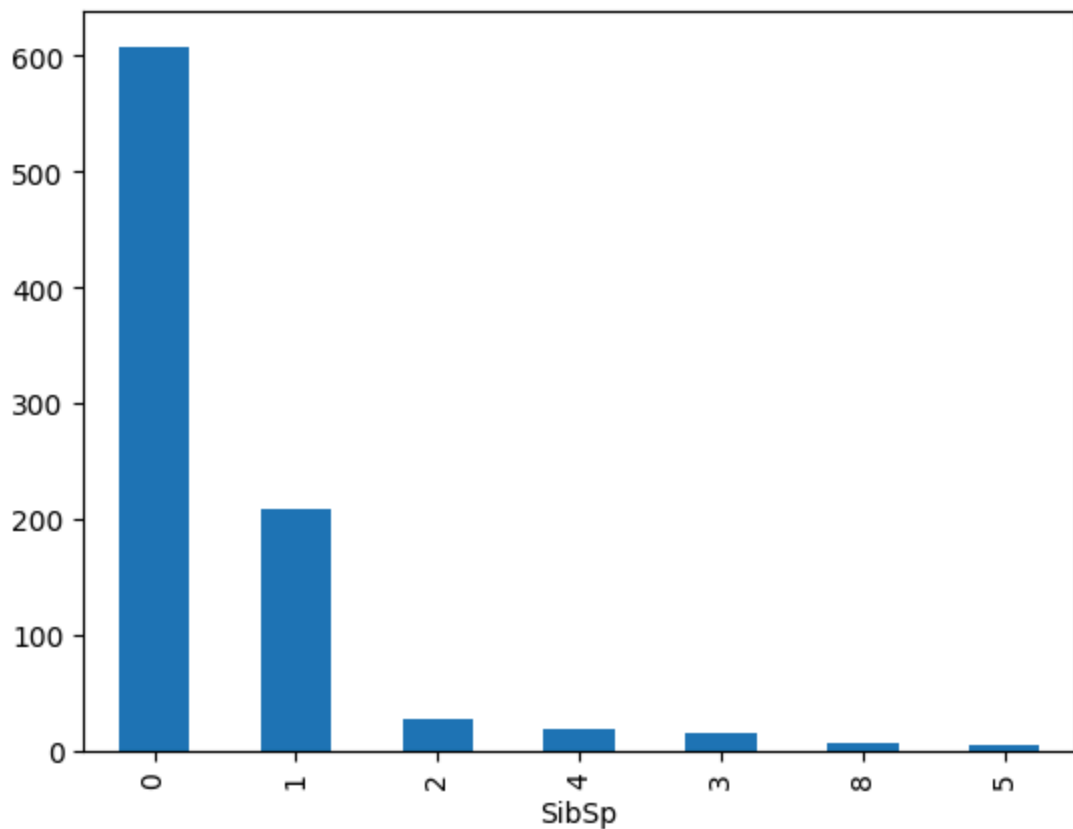
Conclusion

- There are 608 peoples are traveling alone.

```
In [29... titanic['SibSp'].value_counts()
```

```
Out[29]: SibSp
0      608
1      209
2       28
4       18
3       16
8        7
5         5
Name: count, dtype: int64
```

```
In [30... titanic['SibSp'].value_counts().plot(kind='bar');
```



```
In [31]: titanic['SibSp'].isnull().sum()
```

```
Out[31]: np.int64(0)
```

Parch

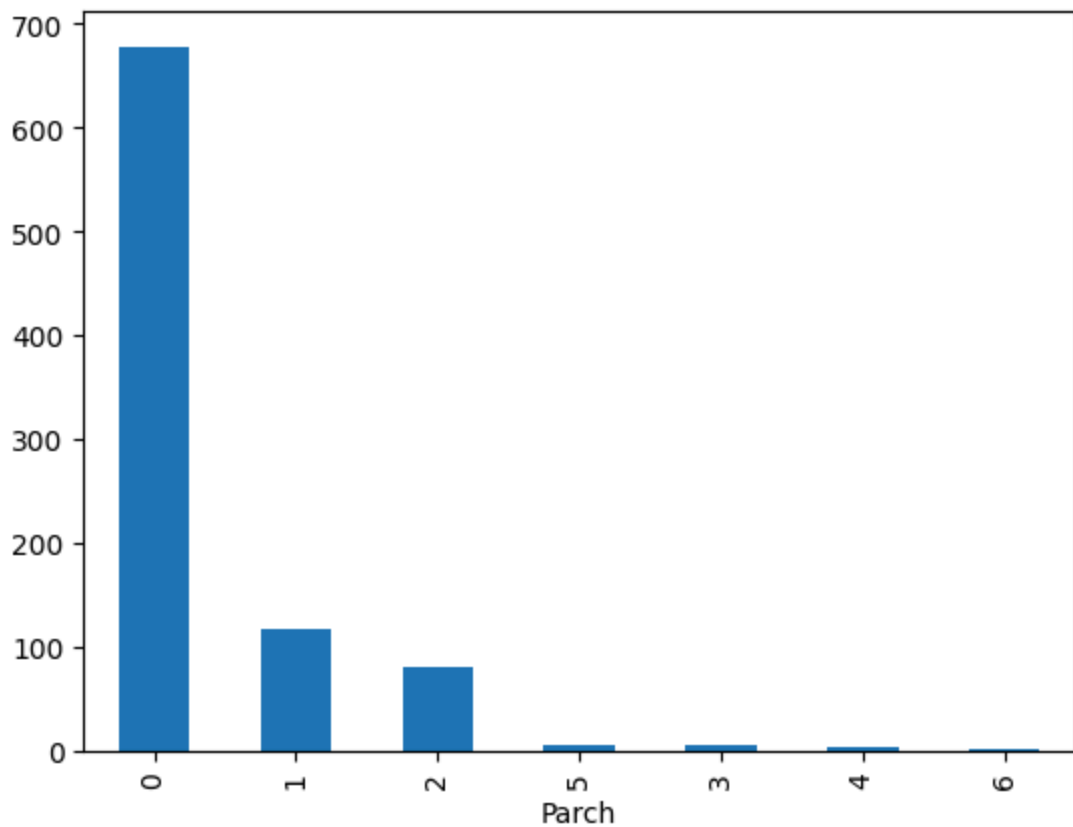
Conclusion

- There are 678 travelers who travel without parents.
- Parch and SibSp cols can be merged to form a new col called family_size.
- Create new col called is_alone.

```
In [32]: titanic['Parch'].value_counts()
```

```
Out[32]: Parch
0      678
1      118
2       80
5        5
3        5
4        4
6        1
Name: count, dtype: int64
```

```
In [33]: titanic['Parch'].value_counts().plot(kind='bar');
```



```
In [34...] titanic['Parch'].isnull().sum()
```

```
Out[34]: np.int64(0)
```

Embarked

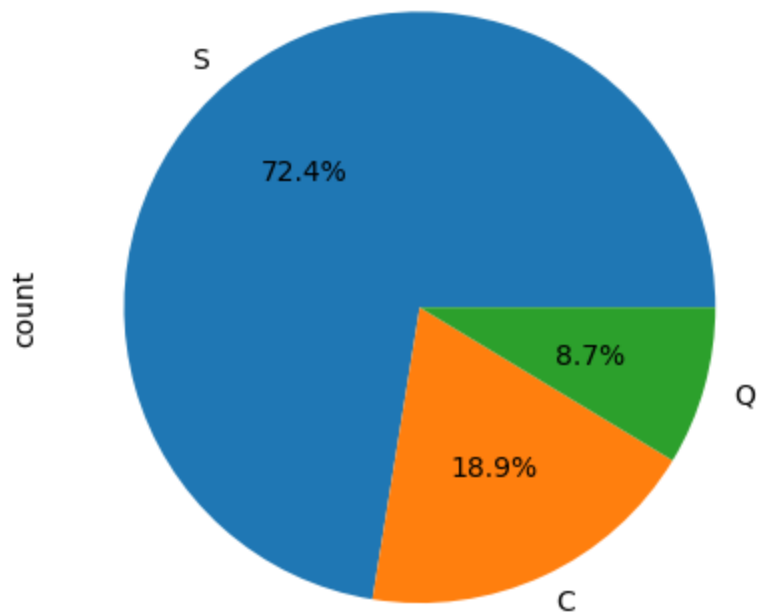
Conclusion

- There are 2 missing values.
- There are 72.4% travelers are from the 'S' and 18.9% from 'C' and 8.7% from the 'Q'.

```
In [35...] titanic['Embarked'].value_counts()
```

```
Out[35]: Embarked
S      644
C      168
Q       77
Name: count, dtype: int64
```

```
In [36...] titanic['Embarked'].value_counts().plot(kind='pie', autopct='%0.1f%%');
```



```
In [37... titanic['Embarked'].isnull().sum()
```

```
Out[37]: np.int64(2)
```

```
In [38... # check the null cols  
titanic[titanic['Embarked'] == None]
```

```
Out[38]: PassengerId  Survived  Pclass  Name  Sex  Age  SibSp  Parch  Ticket  Fare  Cabin  Embarked
```

4. Bivariate Analysis

Bivariate analysis examines the relationship between two variables.

Since `Survived` is our target variable, we will focus on understanding how different passenger characteristics are associated with survival.

We will investigate relationships between:

- Survival and passenger class
- Survival and sex
- Survival and embarkation port
- Survival and age
- Other engineered features and survival

Steps in Bivariate Analysis

- Select Two Cols
- Understand type of relationship
 1. Numerical - Numerical → Plots graph like scatterplot, regression plot, 2D histplot, 2D kdeplot.
 2. Numerical - Categorical → Plots graph like barplot, kdeplot, violinplot, also scatter plot.
 3. Categorical - Categorical → Plots graph like heatmap, stacked barplot, treemaps.
- Conclusions

```
In [39]: titanic.head()
```

```
Out[39]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

4.1 Survival vs Passenger Class

Conclusions

- In Pclass 1 63.0% peoples are survived.
- In Pclass 2 47.3% peoples are survived.
- In Pclass 3 24.2% peoples are survived.
- So here we clear see this in the PClass 3 have highest survived

chance and in the PClass 1 have lowest survived chance.

```
In [40...] pd.crosstab(titanic['Survived'], titanic['Pclass'])
```

```
Out[40]:
```

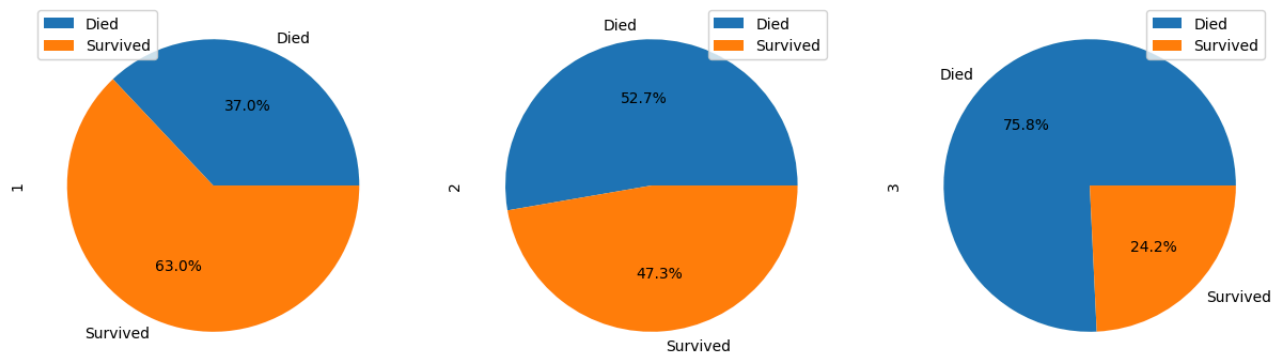
	Pclass	1	2	3
Survived				
0	80	97	372	
1	136	87	119	

```
In [41...] round(pd.crosstab(titanic['Survived'], titanic['Pclass'], normalize='colu
```

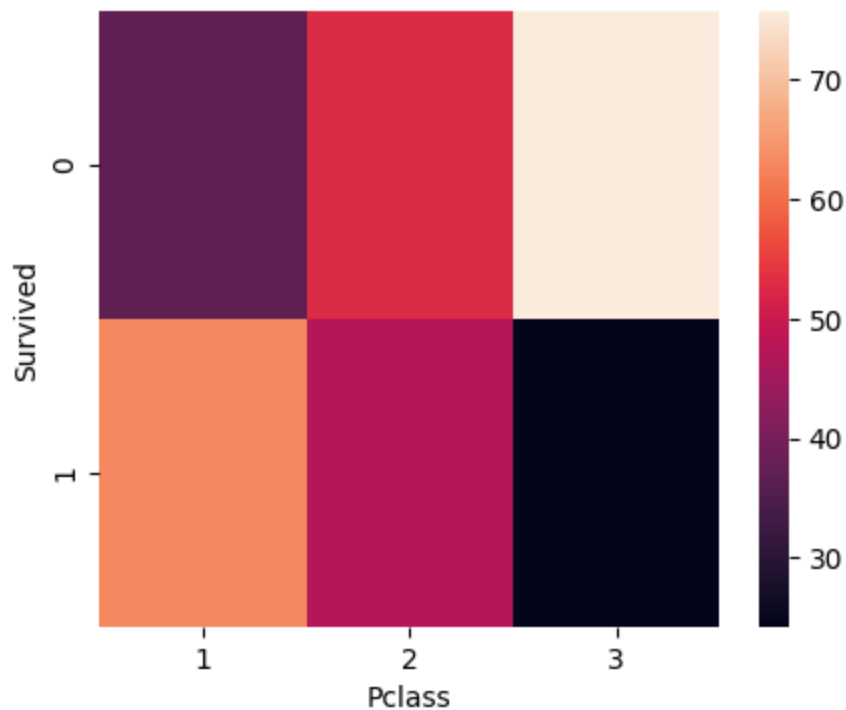
```
Out[41]:
```

	Pclass	1	2	3
Survived				
0	37.04	52.72	75.76	
1	62.96	47.28	24.24	

```
In [42...] round(pd.crosstab(titanic['Survived'], titanic['Pclass'], normalize='colu
```



```
In [43...] plt.figure(figsize=(5,4))
sns.heatmap(round(pd.crosstab(titanic['Survived'], titanic['Pclass'], nor
```



4.2 Survival vs Sex

Conclusions

- There are 74.2% females are survived and only 25.8% are died.
- There are 18.9% males are survived and only 81.1% are died.
- This indicates that Sex is strongly associated with the survival outcome.

```
In [44... round(pd.crosstab(titanic['Survived'], titanic['Sex'], normalize='columns'
```

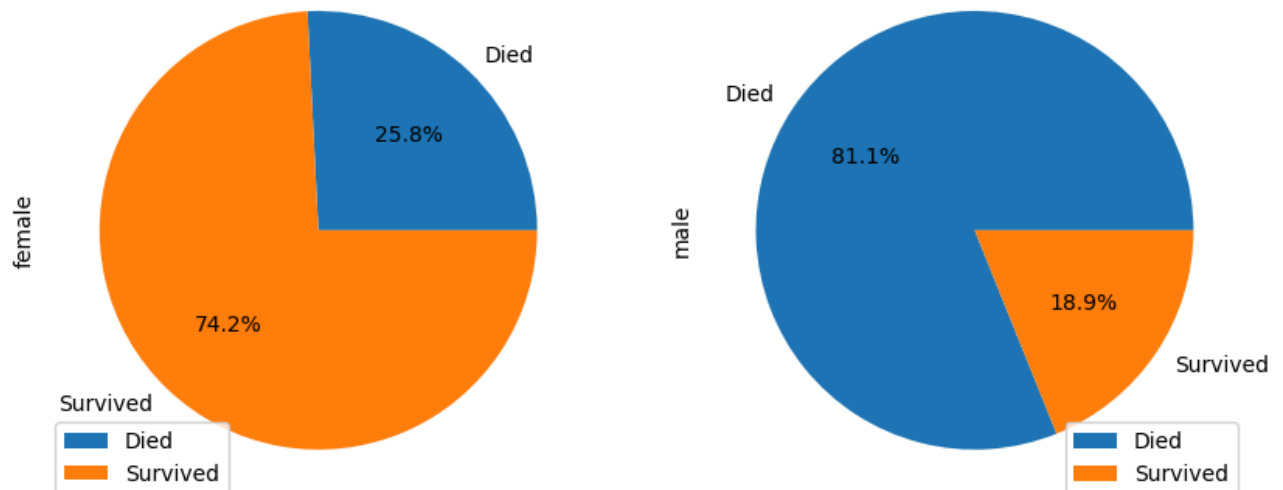
```
Out[44]:      Sex  female  male
```

```
Survived
```

```
0      25.8  81.11
```

```
1      74.2  18.89
```

```
In [45... (pd.crosstab(titanic['Survived'], titanic['Sex'], normalize='columns') *
```



4.3 Survival vs Embarked

Conclusions

- Passengers embarking at C show a higher survival proportion. This may be related to the passenger-class composition of the group, so we examine Pclass and Embarked together.

```
In [46]: round(pd.crosstab(titanic['Survived'], titanic['Embarked'], normalize='co
```

```
Out[46]:
```

Embarked	C	Q	S
Survived			
0	44.64	61.04	66.3
1	55.36	38.96	33.7

Here above we see this that in Embarked C, have survived change is 55.36%, so here may be in Embarked C have most female or may be most passenger travel in Pclass 1.

```
In [47]: # check for female
round(pd.crosstab(titanic['Sex'], titanic['Embarked'], normalize='columns
```

```
Out[47]:
```

Embarked	C	Q	S
Sex			
female	43.45	46.75	31.52
male	56.55	53.25	68.48

Here we see this there our female assumption is not correct

```
In [48... # check for PClass
round(pd.crosstab(titanic['Pclass'], titanic['Embarked'], normalize='colu
```

```
Out[48]:
```

	Embarked	C	Q	S
Pclass				
1		50.60	2.60	19.72
2		10.12	3.90	25.47
3		39.29	93.51	54.81

So we clear see in the Embarked C have most passenger that travel in the Pclass 1.

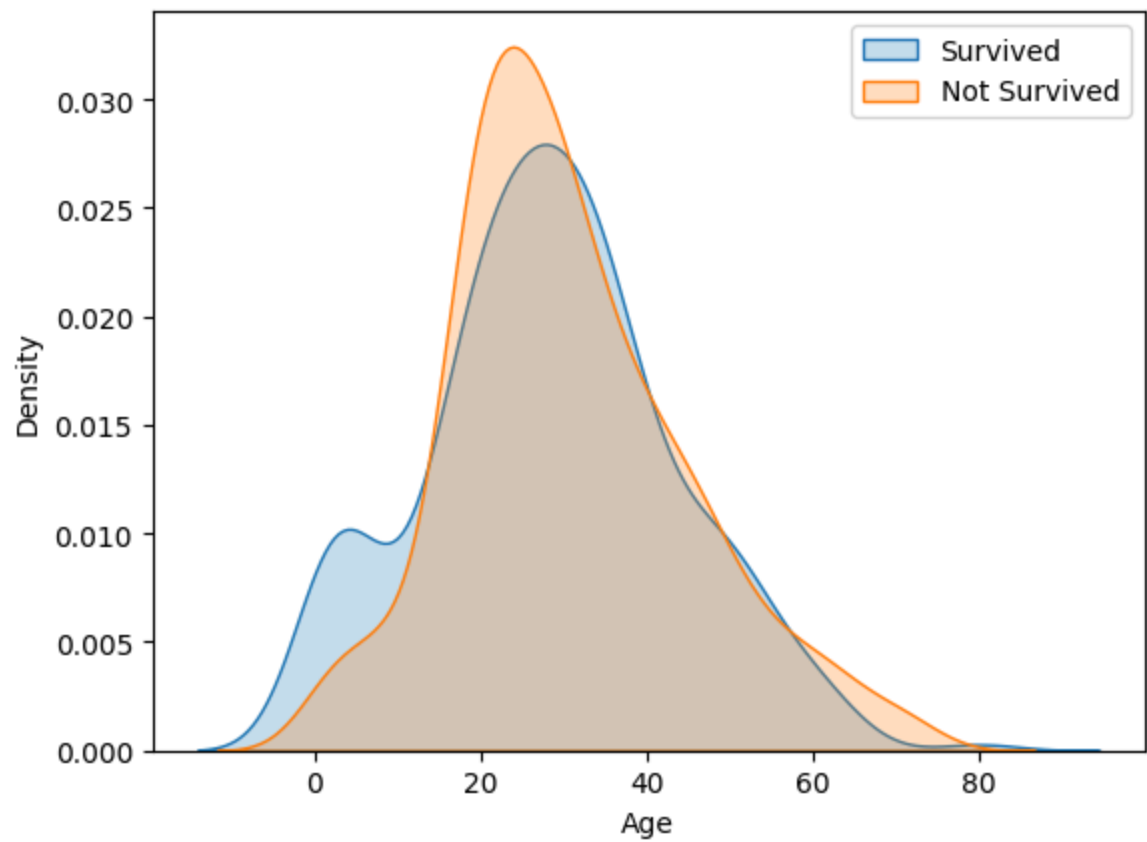
4.4 Survival vs Age

Conclusions

- Give to priority save the kids means 0 to 10 age.
- Also their are high chances of survived if age is 30 - 40, because in this range mostly passenge travel in PClass 1.
- Younger passengers show a different survival distribution compared with older passengers. However, age is also related to other variables such as passenger class, so the observed relationship should not be interpreted as a causal effect of age alone.

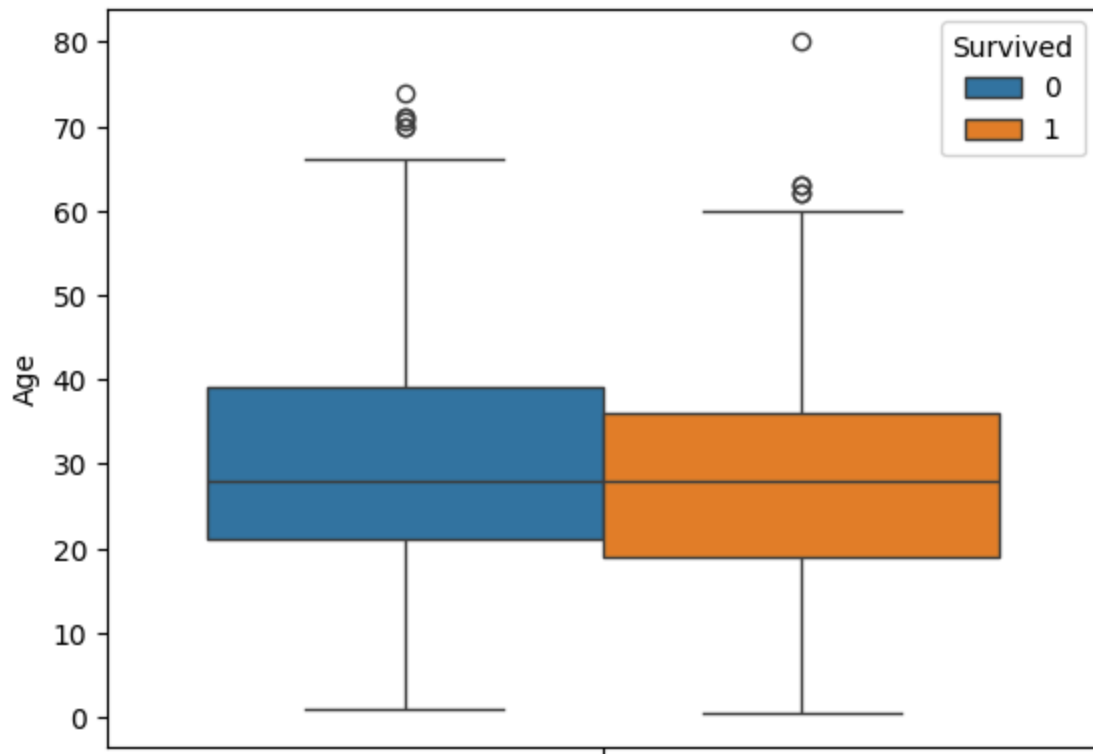
```
In [49... sns.kdeplot(titanic[titanic['Survived'] == 1]['Age'], label='Survived', f
sns.kdeplot(titanic[titanic['Survived'] == 0]['Age'], label='Not Survived
plt.legend()
```

```
Out[49]: <matplotlib.legend.Legend at 0x242f32d8190>
```



```
In [50...] print(titanic[titanic['Pclass'] == 1]['Age'].mean())  
38.233440860215055
```

```
In [51...] sns.boxplot(data=titanic, y='Age', hue='Survived');
```



5.Feature Engineering

5.1 On the Fare Column

The `Fare` column represents the fare paid for a ticket. However, when multiple passengers share the same ticket, the recorded fare can represent the fare associated with the group rather than a single passenger.

To better understand the fare paid per passenger, we create a new feature called `Individual Fare`.

Individual Fare = Fare / Number of passengers sharing the ticket

```
In [52]: titanic['SibSp'].value_counts()
```

```
Out[52]: SibSp
0      608
1      209
2       28
4       18
3       16
8        7
5         5
Name: count, dtype: int64
```

```
In [53]: titanic[titanic['SibSp'] == 8]
```

Out[53]:	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
	159	0	3	Sage, Master. Thomas Henry	male	NaN	8	2	CA. 2343	69.55	NaN
	180	0	3	Sage, Miss. Constance Gladys	female	NaN	8	2	CA. 2343	69.55	NaN
	201	0	3	Sage, Mr. Frederick	male	NaN	8	2	CA. 2343	69.55	NaN
	324	0	3	Sage, Mr. George John Jr	male	NaN	8	2	CA. 2343	69.55	NaN
	792	0	3	Sage, Miss. Stella Anna	female	NaN	8	2	CA. 2343	69.55	NaN
	846	0	3	Sage, Mr. Douglas Bullen	male	NaN	8	2	CA. 2343	69.55	NaN
	863	0	3	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	2	CA. 2343	69.55	NaN

```
In [54]: titanic[titanic['Ticket'] == 'CA. 2343']
```

Out[54]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
159	160	0	3	Sage, Master. Thomas Henry	male	NaN	8	2	CA. 2343	69.55	NaN
180	181	0	3	Sage, Miss. Constance Gladys	female	NaN	8	2	CA. 2343	69.55	NaN
201	202	0	3	Sage, Mr. Frederick	male	NaN	8	2	CA. 2343	69.55	NaN
324	325	0	3	Sage, Mr. George John Jr	male	NaN	8	2	CA. 2343	69.55	NaN
792	793	0	3	Sage, Miss. Stella Anna	female	NaN	8	2	CA. 2343	69.55	NaN
846	847	0	3	Sage, Mr. Douglas Bullen	male	NaN	8	2	CA. 2343	69.55	NaN
863	864	0	3	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	2	CA. 2343	69.55	NaN

Why combine the training and test datasets?

- The `Ticket` feature can represent multiple passengers traveling under the same ticket. Therefore, to understand shared-ticket patterns and calculate individual fare correctly, we temporarily combine the training and test datasets.
- This combined dataset is used only for exploratory feature construction and understanding ticket-level information.
- The target variable `Survived` remains available only for the training observations.

```
In [55... df = pd.read_csv('Data sets/test.csv')
df.head()
```

Out[55]:

	PassengerId	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	892	3	Kelly, Mr. James	male	34.5	0	0	330911	7.8292	NaN	
1	893	3	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	363272	7.0000	NaN	
2	894	2	Myles, Mr. Thomas Francis	male	62.0	0	0	240276	9.6875	NaN	
3	895	3	Wirz, Mr. Albert	male	27.0	0	0	315154	8.6625	NaN	
4	896	3	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	3101298	12.2875	NaN	

```
In [56... titanic = pd.concat([titanic, df])
titanic
```

Out[56]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1.0	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1.0	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0.0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500
...	...	...	...	...	...	...	...	...	...	...
413	1305	NaN	3	Spector, Mr. Woolf	male	NaN	0	0	A.5. 3236	8.0500
414	1306	NaN	1	Oliva y Ocana, Dona. Fermina	female	39.0	0	0	PC 17758	108.9000
415	1307	NaN	3	Saether, Mr. Simon Sivertsen	male	38.5	0	0	SOTON/O.Q. 3101262	7.2500
416	1308	NaN	3	Ware, Mr. Frederick	male	NaN	0	0	359309	8.0500
417	1309	NaN	3	Peter, Master. Michael J	male	NaN	1	1	2668	22.3583

1309 rows × 12 columns

```
In [57... titanic[titanic['Ticket'] == 'CA. 2343']
```

Out[57]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
159	160	0.0	3	Sage, Master. Thomas Henry	male	NaN	8	2	CA. 2343	69.55	N/A
180	181	0.0	3	Sage, Miss. Constance Gladys	female	NaN	8	2	CA. 2343	69.55	N/A
201	202	0.0	3	Sage, Mr. Frederick	male	NaN	8	2	CA. 2343	69.55	N/A
324	325	0.0	3	Sage, Mr. George John Jr	male	NaN	8	2	CA. 2343	69.55	N/A
792	793	0.0	3	Sage, Miss. Stella Anna	female	NaN	8	2	CA. 2343	69.55	N/A
846	847	0.0	3	Sage, Mr. Douglas Bullen	male	NaN	8	2	CA. 2343	69.55	N/A
863	864	0.0	3	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	2	CA. 2343	69.55	N/A
188	1080	NaN	3	Sage, Miss. Ada	female	NaN	8	2	CA. 2343	69.55	N/A
342	1234	NaN	3	Sage, Mr. John George	male	NaN	1	9	CA. 2343	69.55	N/A
360	1252	NaN	3	Sage, Master. William Henry	male	14.5	8	2	CA. 2343	69.55	N/A
365	1257	NaN	3	Sage, Mrs. John (Annie Bullen)	female	NaN	1	9	CA. 2343	69.55	N/A

In [58...

`titanic['Fare']/((titanic['SibSp'] + titanic['Parch'] + 1))`

```
Out[58]: 0      3.625000
          1     35.641650
          2      7.925000
          3     26.550000
          4      8.050000
          ...
          413     8.050000
          414    108.900000
          415      7.250000
          416     8.050000
          417      7.452767
          Length: 1309, dtype: float64
```

```
In [59... titanic['Individual Fare'] = titanic['Fare']/(titanic['SibSp'] + titanic['Parch'])
```

```
In [60... titanic.head()
```

```
Out[60]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Individual Fare
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	3.625000
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence Briggs Th...)	female	38.0	1	0	PC 17599	71.2833	108.900000
2	3	1.0	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	7.925000
3	4	1.0	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	26.550000
4	5	0.0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	8.050000

Univariate Analysis on Individual Fare

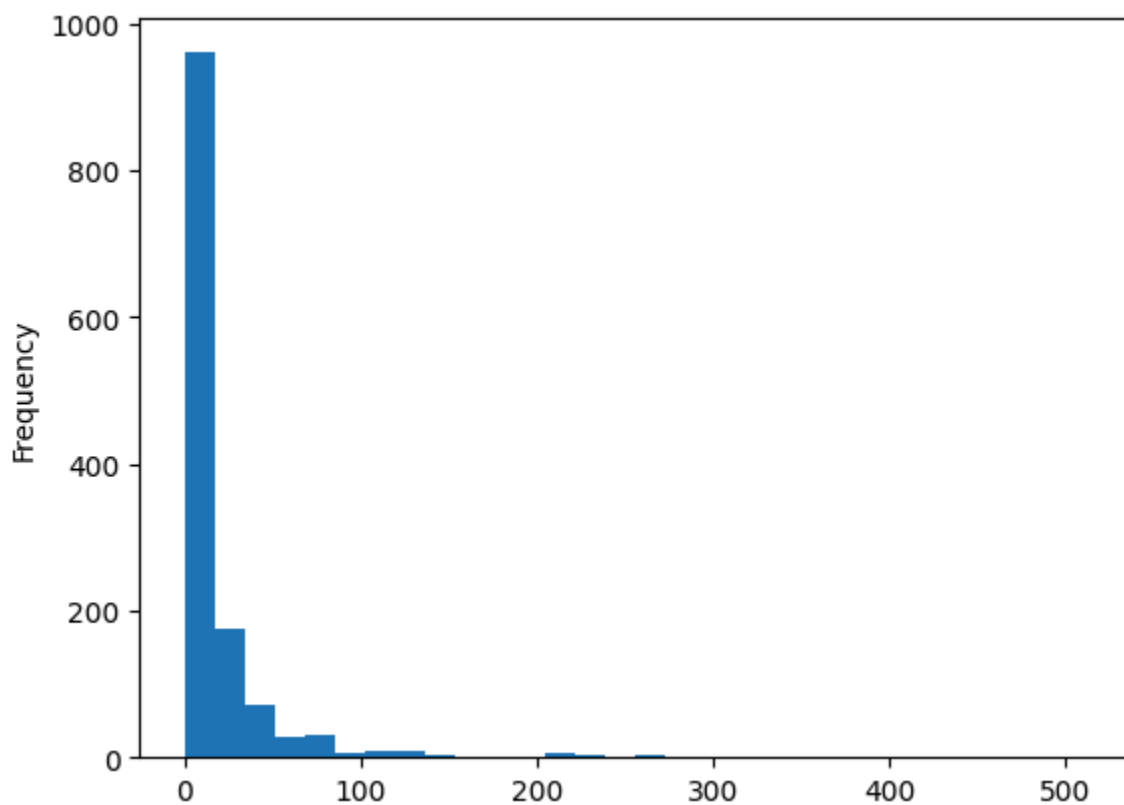
Colclusion

- There are one male and a female their ticket price is 512\$, which is suprisely.

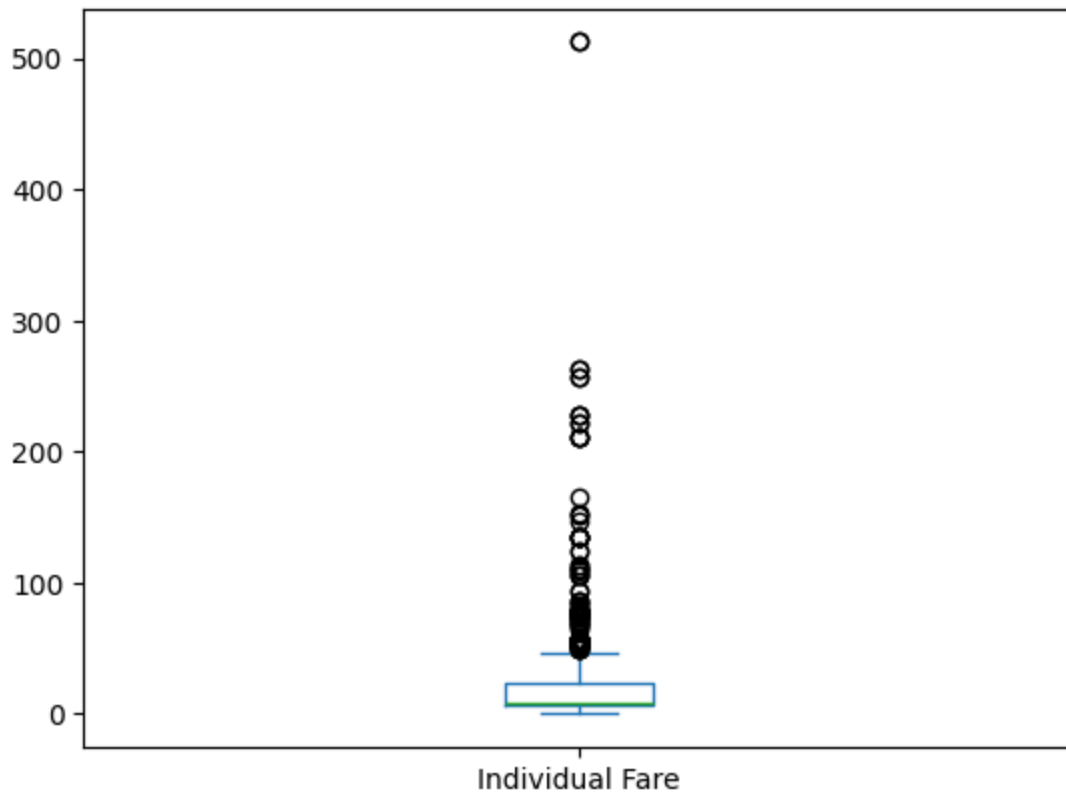
```
In [61... titanic['Individual Fare'].describe()
```

```
Out[61]: count    1308.000000
         mean      20.518215
         std       35.774337
         min        0.000000
         25%        7.452767
         50%        8.512483
         75%       24.237500
         max       512.329200
         Name: Individual Fare, dtype: float64
```

```
In [62]: titanic['Individual Fare'].plot(kind='hist', bins=30);
```



```
In [63]: titanic['Individual Fare'].plot(kind='box');
```



```
In [64]: titanic[titanic['Individual Fare'] > 500]
```

Out[64]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Ca
258	259	1.0	1	Ward, Miss. Anna	female	35.0	0	0	PC 17755	512.3292	
737	738	1.0	1	Lesurer, Mr. Gustave J	male	35.0	0	0	PC 17755	512.3292	B

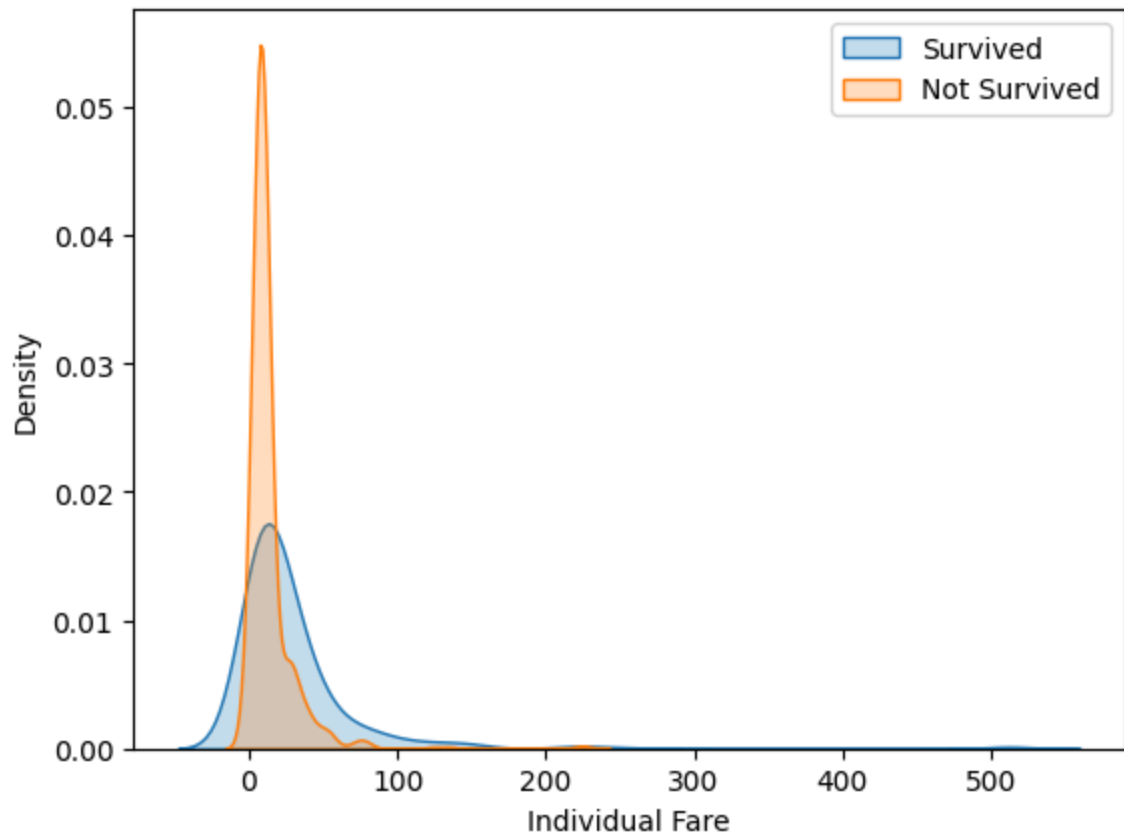
Bivariate Analysis on Individual Fare

Colclusion

- People who paid higher fare the survived chance is more.

```
In [65]: sns.kdeplot(titanic[titanic['Survived'] == 1]['Individual Fare'], label='Survived=1')
sns.kdeplot(titanic[titanic['Survived'] == 0]['Individual Fare'], label='Survived=0')
plt.legend()
```

Out[65]: <matplotlib.legend.Legend at 0x242f33125d0>



5.2 On the SibSp and Parch column

`SibSp` and `Parch` both provide information about a passenger's family relationships aboard the Titanic.

We combine these variables to create:

Family Size = SibSp + Parch + 1

The additional `1` represents the passenger themselves.

```
In [66... # create 'Family_size' col
titanic['Family_size'] = titanic['Parch'] + titanic['SibSp'] + 1
```

```
In [67... titanic.head()
```

Out[67]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	
2	3	1.0	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1.0	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C
4	5	0.0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

In [68...

```
# create 'Family type' col
# 1 = alone
# 2 to 4 = small
# > 5 = large

def family_size(num):
    if num == 1:
        return 'Alone'
    elif num > 1 and num < 5:
        return 'Small'
    else:
        return 'Large'
```

In [69...

```
titanic['Family size'].apply(family_size)
```

Out[69]:

0 Small
1 Small
2 Alone
3 Small
4 Alone
...
413 Alone
414 Alone
415 Alone
416 Alone
417 Small
Name: Family size, Length: 1309, dtype: object

```
In [70]: titanic['Family Type'] = titanic['Family size'].apply(family_size)
```

```
In [71]: titanic.head()
```

```
Out[71]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	
2	3	1.0	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1.0	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C
4	5	0.0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

Univariate Analysis on Family size and Family Type

Colclusion

- Most of the peoples are traveling with 1 to 4 memebers.
- There are a family with traveling with 11 members in Pclass 3.
- There are 60.4% peoples are traving Alone.

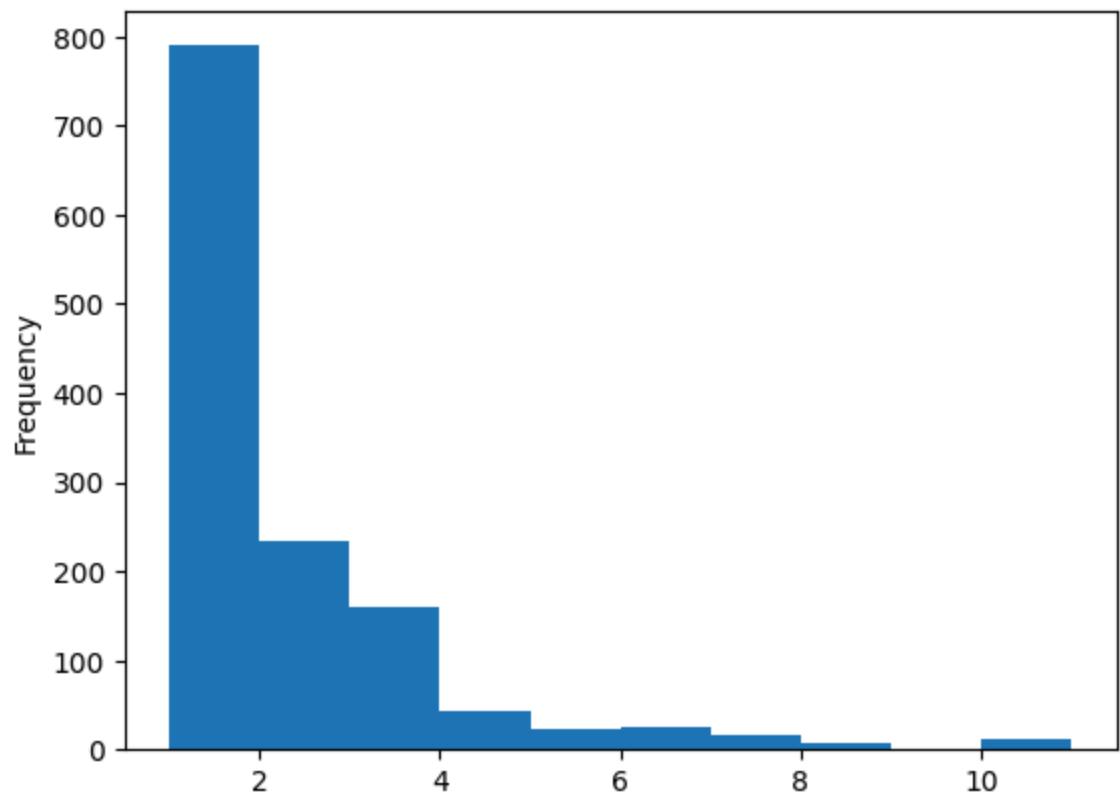
```
In [72]: # Family size
titanic['Family size'].describe()
```

```
Out[72]:
```

count	1309.000000
mean	1.883881
std	1.583639
min	1.000000
25%	1.000000
50%	1.000000
75%	2.000000
max	11.000000

Name: Family size, dtype: float64

```
In [73]: titanic['Family size'].plot(kind='hist');
```



In [74... `titanic[titanic['Family size'] == 11]`

Out[74]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
159	160	0.0	3	Sage, Master. Thomas Henry	male	NaN	8	2	CA. 2343	69.55	N/A
180	181	0.0	3	Sage, Miss. Constance Gladys	female	NaN	8	2	CA. 2343	69.55	N/A
201	202	0.0	3	Sage, Mr. Frederick	male	NaN	8	2	CA. 2343	69.55	N/A
324	325	0.0	3	Sage, Mr. George John Jr	male	NaN	8	2	CA. 2343	69.55	N/A
792	793	0.0	3	Sage, Miss. Stella Anna	female	NaN	8	2	CA. 2343	69.55	N/A
846	847	0.0	3	Sage, Mr. Douglas Bullen	male	NaN	8	2	CA. 2343	69.55	N/A
863	864	0.0	3	Sage, Miss. Dorothy Edith "Dolly"	female	NaN	8	2	CA. 2343	69.55	N/A
188	1080	NaN	3	Sage, Miss. Ada	female	NaN	8	2	CA. 2343	69.55	N/A
342	1234	NaN	3	Sage, Mr. John George	male	NaN	1	9	CA. 2343	69.55	N/A
360	1252	NaN	3	Sage, Master. William Henry	male	14.5	8	2	CA. 2343	69.55	N/A
365	1257	NaN	3	Sage, Mrs. John (Annie Bullen)	female	NaN	1	9	CA. 2343	69.55	N/A

In [75...

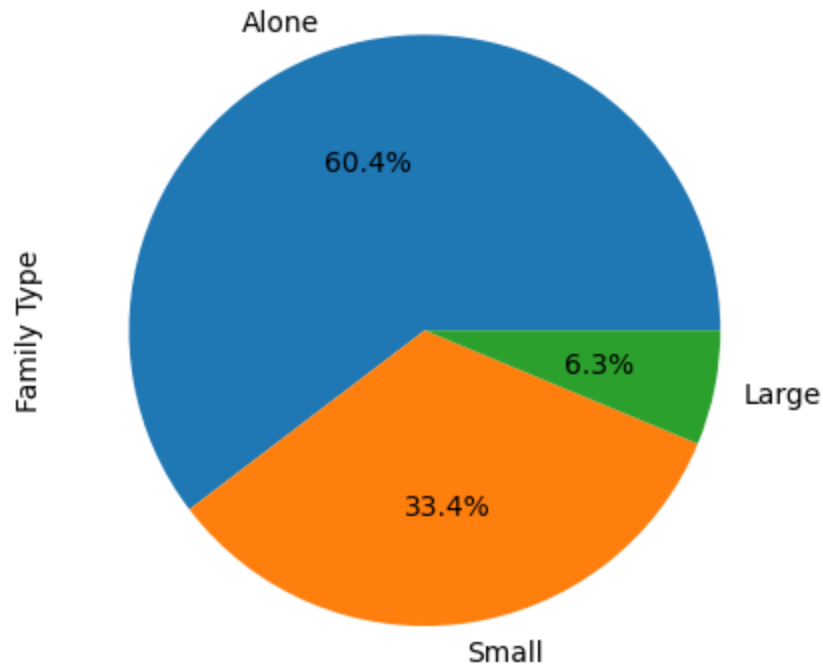
```
# Family type
titanic['Family Type'].value_counts()
```

Out[75]:

```
Family Type
Alone      790
Small     437
Large       82
Name: count, dtype: int64
```

In [76...

```
titanic['Family Type'].value_counts().plot(kind='pie', autopct='%0.1f%%',
```



Bivariate Analysis on Family Type

Colclusion

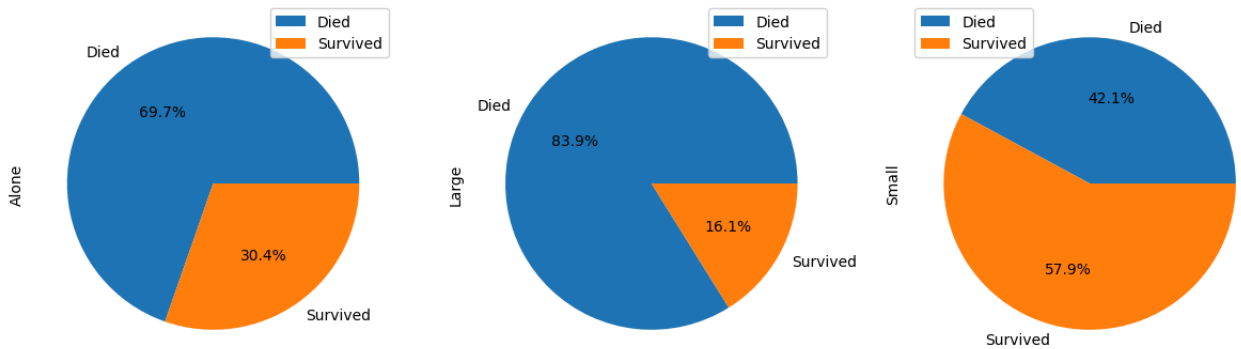
- In Small family the % of servived people is 57.9%.
- In Alone family the % of servived people is 30.1%.
- In Large family the % of servived people is 16.1%.

```
In [77...] round(pd.crosstab(titanic['Survived'], titanic['Family Type'], normalize=
```

```
Out[77]: Family Type Alone Large Small
```

Survived				
	0.0	69.65	83.87	42.12
	1.0	30.35	16.13	57.88

```
In [78...] round(pd.crosstab(titanic['Survived'], titanic['Family Type'], normalize=
```



5.3 On the Name column

Extracting Information from Name

The `Name` column contains structured information that can be extracted into useful features.

We create:

- `Surname` — family surname
- `Title` — social/professional title extracted from the passenger's name

Rare titles are grouped into `Other` to reduce the number of categories.

```
In [79... # make Surname col
titanic['Surname'] = titanic['Name'].str.split(',').str.get(0)
```

```
In [80... # create new col title
titanic['tittle'] = titanic['Name'].str.split(',').str.get(1).str.strip()
```

```
In [81... titanic['tittle'].value_counts()
```

```
Out[81]: tittle
Mr.      757
Miss.    260
Mrs.     197
Master.   61
Rev.       8
Dr.        8
Col.       4
Major.     2
Mlle.      2
Ms.        2
Mme.       1
Don.       1
Sir.       1
Lady.      1
Capt.     1
the        1
Jonkheer.  1
Dona.      1
Name: count, dtype: int64
```

```
In [82]: # replace the unknown title into other
replacement = {
    'Rev.': 'Other',
    'Dr.': 'Other',
    'Col.': 'Other',
    'the': 'Other',
    'Major.': 'Other',
    'Capt.': 'Other',
    'Jonkheer.': 'Other',
    'Lady.': 'Mrs.',
    'Mme.': 'Mrs.',
    'Ms.': 'Mrs.',
    'Mlle.': 'Mrs.',
    'Dona.': 'Mrs.',
    'Sir.': 'Mr.',
    'Don.': 'Other'
}
```

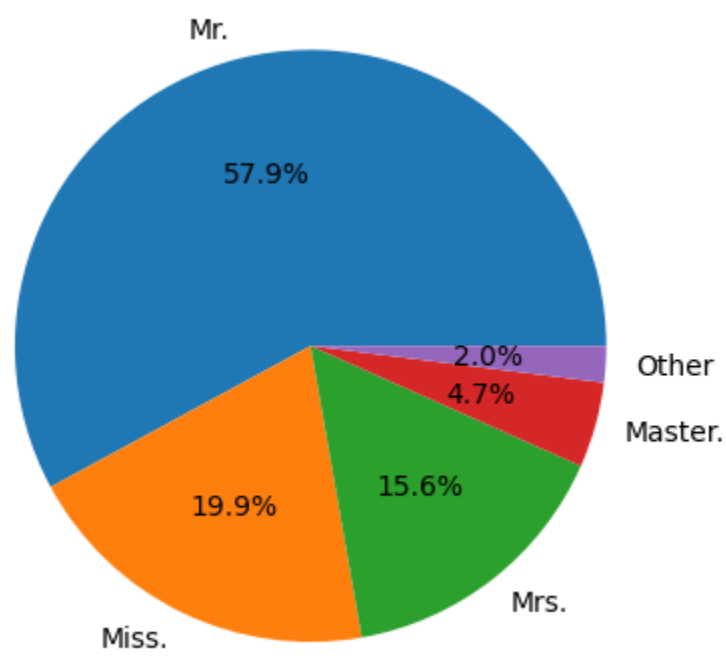
```
In [83]: titanic['tittle'] = titanic['tittle'].replace(replacement)
```

Univariate Analysis on 'tittle' col

```
In [84]: titanic['tittle'].value_counts()
```

```
Out[84]: tittle
Mr.      758
Miss.    260
Mrs.     204
Master.   61
Other     26
Name: count, dtype: int64
```

```
In [85... titanic['tittle'].value_counts().plot(kind='pie', autopct='%0.01f%%', yla
```



Bivariate Analysis on 'tittle' col

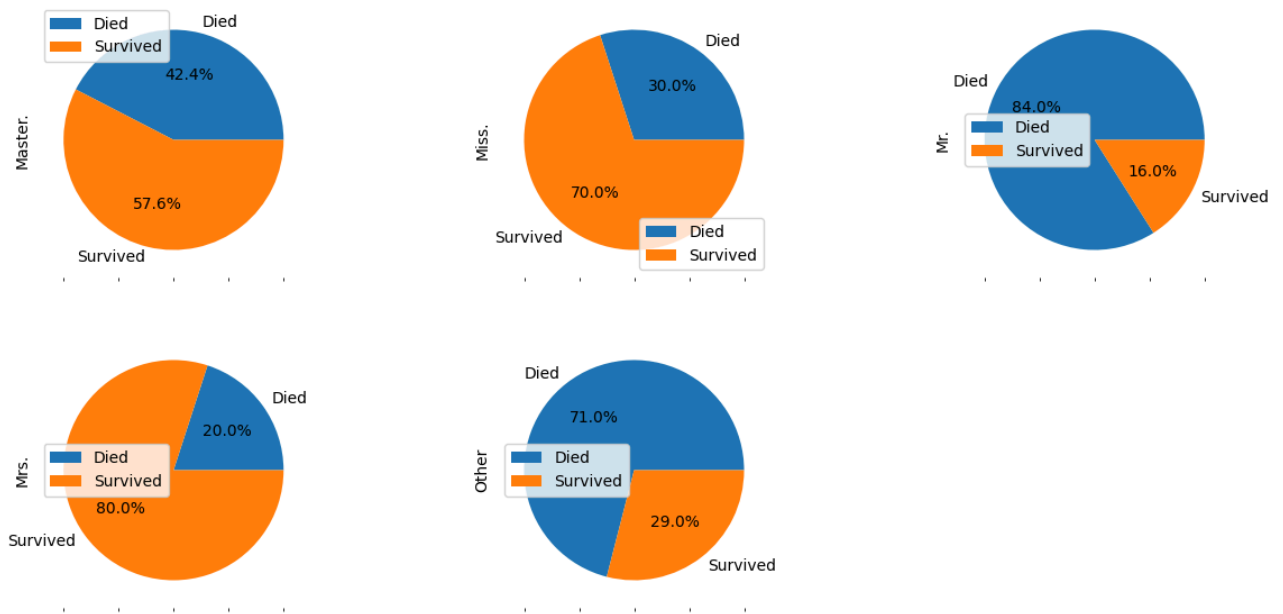
```
In [86... temp_df = round(pd.crosstab(titanic['Survived'], titanic['tittle'], norma
```

```
In [87... temp_df
```

Out[87]:

tittle	Master.	Miss.	Mr.	Mrs.	Other
Survived					
0.0	42.0	30.0	84.0	20.0	71.0
1.0	57.0	70.0	16.0	80.0	29.0

```
In [88... temp_df.plot(kind='pie', subplots=True, figsize=(15,7),layout=(2, 3), aut
```



In [89... temp_df

Out[89]:

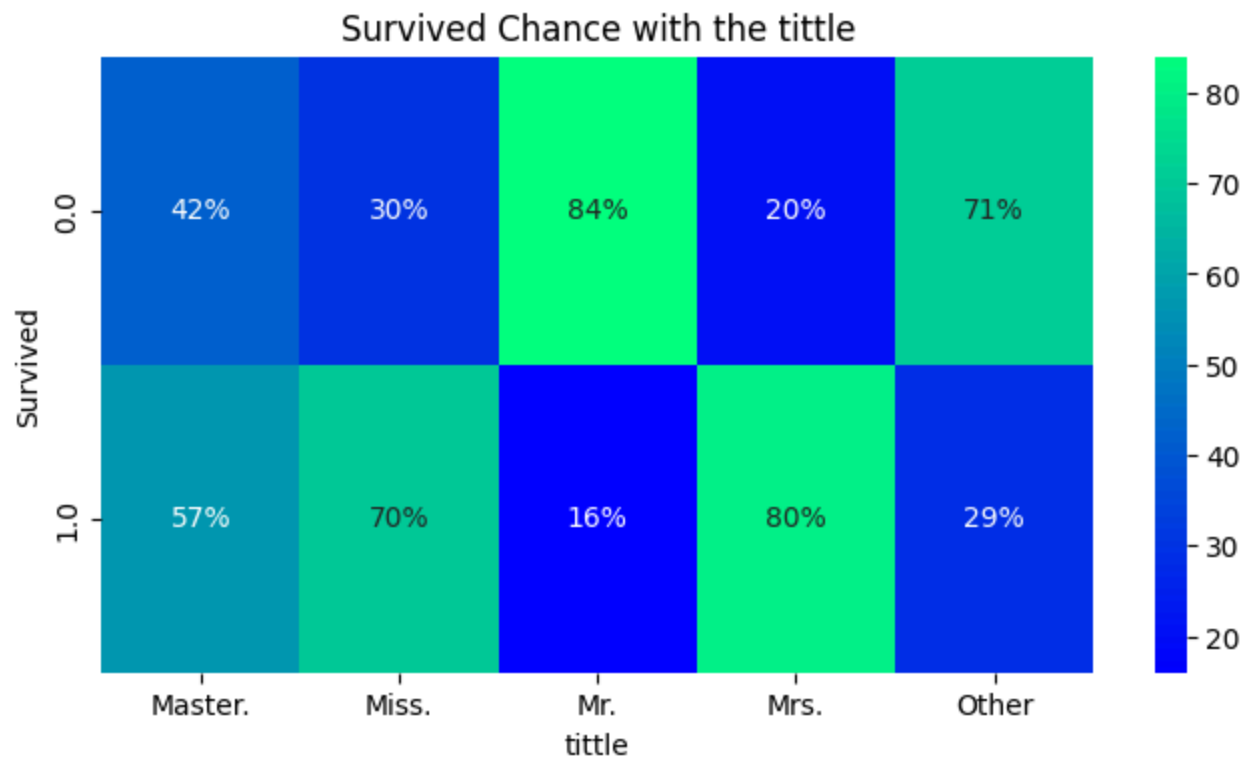
	title	Master.	Miss.	Mr.	Mrs.	Other
	Survived					
0.0		42.0	30.0	84.0	20.0	71.0
1.0		57.0	70.0	16.0	80.0	29.0

```
In [90... plt.figure(figsize=(8,4))
sns.heatmap(temp_df, annot=True, cmap='winter')

# add % sign
for text in plt.gca().texts:
    text.set_text(text.get_text() + '%')

plt.title('Survived Chance with the tittle')
```

Out[90]: Text(0.5, 1.0, 'Survived Chance with the tittle')



5.4 On the Cabin column

Extracting Deck from Cabin

- The `Cabin` column contains a large number of missing values.
- Instead of using the complete cabin number directly, we extract the first character of the cabin value, which represents the deck.
- For passengers without a recorded cabin, we assign `None`.
- This creates a simpler categorical feature called `Deck`.

```
In [91...] print(f'{round(titanic['Cabin'].isnull().sum() / len(titanic['Cabin'])) *  
77%
```

```
In [92...] titanic[titanic['Cabin'] == 'G6' ]
```

Out[92]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
10	11	1.0	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7000
205	206	0.0	3	Strom, Miss. Telma Matilda	female	2.0	0	1	347054	10.4625
251	252	0.0	3	Strom, Mrs. Wilhelm (Elna Matilda Persson)	female	29.0	1	1	347054	10.4625
394	395	1.0	3	Sandstrom, Mrs. Hjalmar (Agnes Charlotta Bengt...	female	24.0	0	2	PP 9549	16.7000
117	1009	NaN	3	Sandstrom, Miss. Beatrice Irene	female	1.0	1	1	PP 9549	16.7000

In [93...

titanic[titanic['Surname'] == 'Sandstrom']

Out[93]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
10	11	1.0	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7	C
394	395	1.0	3	Sandstrom, Mrs. Hjalmar (Agnes Charlotta Bengt...	female	24.0	0	2	PP 9549	16.7	C
117	1009	NaN	3	Sandstrom, Miss. Beatrice Irene	female	1.0	1	1	PP 9549	16.7	C

In [94...

titanic['Cabin']

```
Out[94]: 0      NaN
          1      C85
          2      NaN
          3     C123
          4      NaN
          ...
         413     NaN
         414    C105
         415     NaN
         416     NaN
         417     NaN
          Name: Cabin, Length: 1309, dtype: object
```

```
In [95... titanic['Cabin'].value_counts()
```

```
Out[95]: Cabin
          C23 C25 C27      6
          G6      5
          B57 B59 B63 B66  5
          F33      4
          F4      4
          ...
          C39      1
          B24      1
          D40      1
          D38      1
          C105     1
          Name: count, Length: 186, dtype: int64
```

```
In [96... titanic['Cabin'].fillna('None', inplace=True)
```

```
In [97... titanic['Cabin'].value_counts()
```

```
Out[97]: Cabin
          None      1014
          C23 C25 C27      6
          G6      5
          B57 B59 B63 B66  5
          F33      4
          ...
          C39      1
          B24      1
          D40      1
          D38      1
          C105     1
          Name: count, Length: 187, dtype: int64
```

```
In [98... titanic['Deck'] = titanic['Cabin'].str[0]
```

```
In [99... titanic['Deck'] = titanic['Deck'].str.replace('N', 'None')
```

```
In [10... titanic.head()
```

Out[100]:	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	
2	3	1.0	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1.0	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	
4	5	0.0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

Univarite Analaysis On Deck col

Colclusions

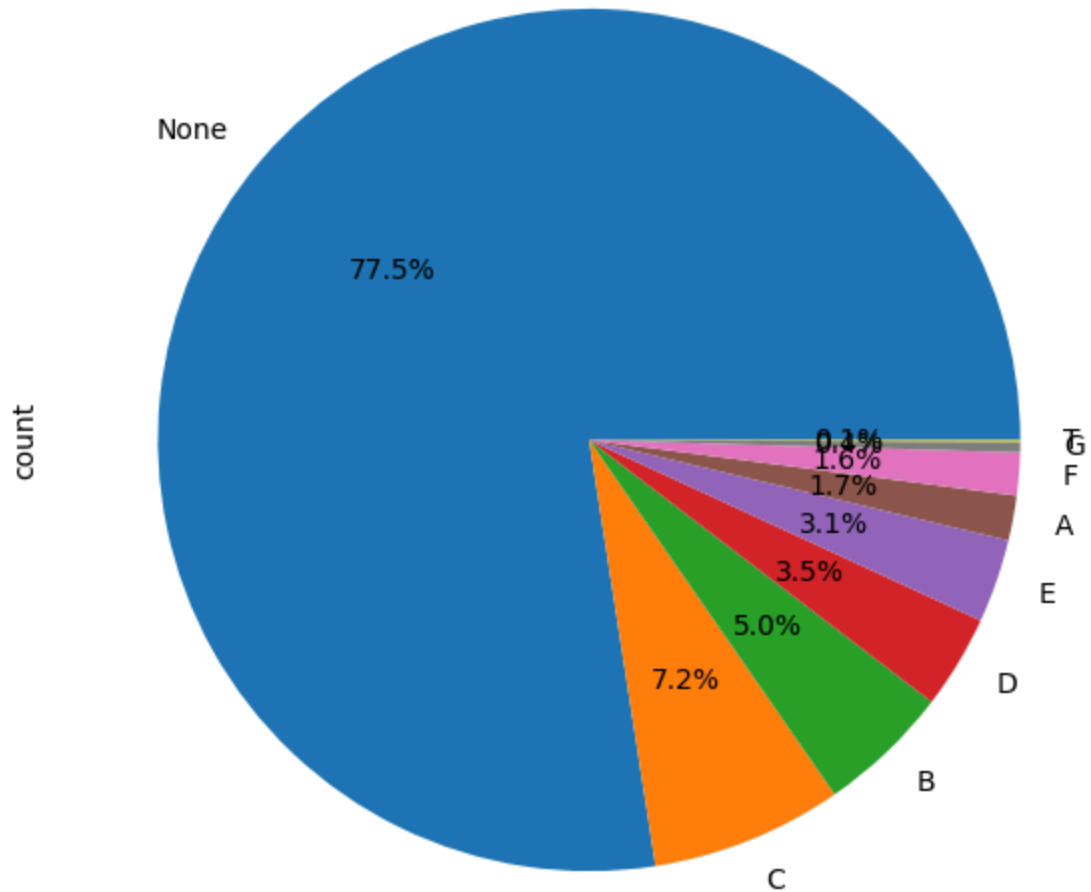
- There are 77.5% value are None.

```
In [10...] titanic['Deck'].value_counts()
```

```
Out[101]: Deck
None      1014
C           94
B           65
D           46
E           41
A           22
F           21
G            5
T            1
Name: count, dtype: int64
```

```
In [10...] titanic['Deck'].value_counts().plot(kind='pie', autopct='%0.1f%%', figsize=
```

```
Out[102]: <Axes: ylabel='count'>
```



Bivariate Analysis On Deck col

Conclusions

- There are most survived chance in the Deck B, D, E, F compared to other Deck.
- All Passengers are died that are in Deck T.
- In Deck G 50% passengers are died.

In [10... `pd.crosstab(titanic['Survived'], titanic['Deck'])`

Out[103]:

Deck	A	B	C	D	E	F	G	None	T
Survived									
0.0	8	12	24	8	8	5	2	481	1
1.0	7	35	35	25	24	8	2	206	0

```
In [10... round(pd.crosstab(titanic['Survived'], titanic['Deck'], normalize='column
```

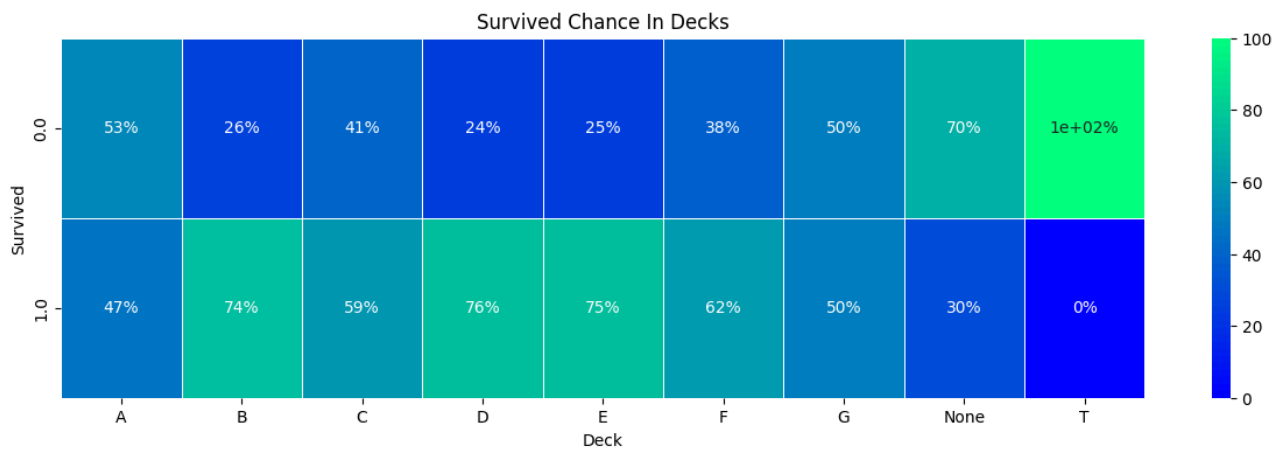
Out[104]:

Deck	A	B	C	D	E	F	G	None	T
Survived									
0.0	53.33	25.53	40.68	24.24	25.0	38.46	50.0	70.01	100.0
1.0	46.67	74.47	59.32	75.76	75.0	61.54	50.0	29.99	0.0

```
In [10... plt.figure(figsize=(15,4))
sns.heatmap(round(pd.crosstab(titanic['Survived'], titanic['Deck'], norma

# add % sign
for text in plt.gca().texts:
    text.set_text(text.get_text() + '%')
plt.title('Survived Chance In Decks')
```

Out[105]: Text(0.5, 1.0, 'Survived Chance In Decks')



```
In [10... titanic.head()
```

Out[106]:	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0.0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1.0	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	
2	3	1.0	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1.0	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	
4	5	0.0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

6. Key Findings

Based on the exploratory analysis, several important patterns were observed:

Passenger Class

- Survival proportion differs considerably across passenger classes, with first-class passengers showing a higher survival proportion than passengers in lower classes.

Sex

- There is a substantial difference in survival proportions between male and female passengers.

Age

- Age distributions differ between survivors and non-survivors, although age is also related to other passenger characteristics.

Embarked

- Survival proportions vary by embarkation port. Passenger-class composition provides one possible factor associated with this difference.

Family Size

- Passengers belonging to small families show a different survival pattern compared with passengers traveling alone or in large families.

Fare

- Fare is strongly skewed, and the analysis suggests that fare-related variables may contain useful information about passenger survival.

Title

- Passenger titles reveal additional demographic and social information that may be useful for further analysis.

Deck

- Deck information shows differences in survival proportions, although the large number of missing cabin values limits interpretation.

7. Next Steps

The EDA provides several features that may be useful for predictive modeling.

Potential next steps include:

- Handling missing values
- Encoding categorical variables
- Selecting relevant features
- Scaling numerical variables where appropriate
- Building classification models
- Evaluating model performance

In []: